LNMIIT
The LNM Institute of Information Technology

The LNM Institute of Information Technology, Jaipur
Software Development Laboratory
Course Code: CS314
Lab Assignment 2: Solution
Project Identification and Initial Project Definition

1. Project Identification

Item	Details
Project Name	Student Wellness Portal
Project Type	Web-based Institutional Wellness, Counseling & Activity Tracking Platform
Team ID	T039
Team Members	Mehul Baid, Alakh Agarwal, Yug Nahar, Daksh Mittal
Roll Numbers	24UCS003, 24UCS001, 24UCS004, 24UCS002
Development Model	Agile + Incremental Development
Primary Users	Students, Wellness Staff / Counselors, Administrators
Development Team	Student software development team
Course	Software Engineering / Software Development Laboratory (CS314)
Project Objective	To develop a centralized platform for managing institutional student wellness, counseling services, habit tracking, and psychoeducation
Expected Outcome	A working, tested and documented software system developed incrementally

2. Project Background

Current Problem

Campus student wellness, mental health support, and lifestyle coaching across university campuses are frequently managed through multiple disconnected mechanisms such as informal emails, physical notice boards, manual register logbooks, or ad-hoc messaging. This fragmentation creates significant hesitation among students seeking assistance due to privacy concerns, perceived social stigma, and lack of visibility into available counseling resources. Counselors lack a streamlined triage workflow, and administrators possess no real-time data on student wellbeing trends or service demand.

Proposed Solution

Student Wellness Portal will provide a centralized, confidential web-based platform for discovering institutional wellness services, submitting structured support requests, scheduling conflict-free counseling appointments, accessing evidence-based psychoeducational resources, and tracking personal wellness habits with comprehensive role-based access control.

3. Project Vision

Student Wellness Portal aims to provide a centralized, user-friendly, confidential, and reliable platform for discovering campus wellness services, requesting psychological and nutritional support, scheduling counseling consultations, accessing educational resources, and tracking personal wellbeing activities in a stigma-free institutional environment.

4. Problem Statement

Campus institutions accommodate diverse student cohorts navigating intense academic coursework, examination anxiety, hostel lifestyle transitions, and emotional distress. Existing institutional wellness practices rely on manual, scattered, and non-confidential channels, resulting in delayed case triage, low service discoverability, scheduling conflicts, and student reluctance to seek timely help. Student Wellness Portal is proposed as an integrated software solution to streamline wellness discovery, service requests, appointment scheduling, psychoeducation, and self-care tracking through a single institutional platform.

5. Project Objectives

The initial objectives should be:

1. To provide a centralized platform for publishing and discovering campus wellness services.
2. To provide a confidential, structured support request submission workflow for students.
3. To allow authorized wellness staff and counselors to triage, manage, and resolve student support cases.
4. To provide a slot-based appointment scheduling mechanism that prevents double-booking and supports Telehealth integration.
5. To maintain a structured digital library of psychoeducational articles, coping guides, and wellness tips.
6. To enable students to log daily self-care habits (sleep, meditation, hydration, workouts) and review visual trend analytics.
7. To provide automated in-app notifications and real-time status tracking for requests and appointments.
8. To provide institutional administrators and management with aggregated, anonymized wellness analytics and incident reports.
9. To maintain appropriate access control and strict data confidentiality based on user roles.
10. To develop the system incrementally using software engineering practices.

6. Stakeholders

Stakeholder	Interest / Role
Students	Discover wellness services, submit confidential support requests, book counseling slots, access self-help resources, and track daily self-care activities.
Wellness Staff / Counselors	Triage support requests, record confidential clinical remarks, manage consultation calendars, and publish educational wellness articles.

Faculty / Department Advisors	Guide students towards campus wellness resources and monitor general academic wellness advisories.
Campus Health Center Staff	Receive and manage acute medical/psychiatric emergency escalations and coordinate clinical referrals.
Campus Administrator	Manage users, assign system roles, configure services, broadcast urgent wellness announcements, and maintain system security.
University / Institute Management	Monitor aggregated student wellbeing trends, evaluate service utilization, and formulate campus welfare policies.
System Development Team	Design, develop, test, document, and maintain the Student Wellness Portal.

7. Initial Scope

Student Wellness Portal will initially cover:

- User registration and secure authentication with institutional email verification (@lnmiit.ac.in)
- Role-based access control (Student, Wellness Staff/Counselor, Administrator)
- Service directory with category filtering, search, and detailed service specifications
- Structured multi-step service request submission with urgency classification (Routine, Priority, Urgent)
- Request status lifecycle tracking (Pending Review, Triaged, Scheduled, In Progress, Completed, Closed)
- Appointment slot scheduling with counselor conflict prevention and Telehealth URL generation
- Psychoeducational resource library with keyword search and category browsing
- Daily habit and mood tracking (sleep, water, meditation, physical exercise) with streak counting
- 7-day and 30-day visual habit trend analytics
- Automated in-app notifications and case timeline audit logs
- Counselor triage queue and confidential consultation remarks logging
- Administrative analytics dashboard and campus broadcast announcements

Initially Out of Scope

To keep the project manageable, we should initially exclude things such as:

- Automated psychiatric medical diagnoses or clinical prescription dispensing
- Electronic Health Record (EHR) integration with external hospital billing systems
- Pharmacy inventory and physical medicine supply tracking
- Continuous biometric hardware/wearable sensor synchronization (e.g., Fitbit/Apple Watch)
- Paid commercial counseling payment gateway processing
- Academic grading and university transcript synchronization
- Complete ERP and hostel room allocation management
- Native video conferencing server hosting (relies on integrated external meeting URLs)

8. Initial Assumptions

- Users will have access to the Internet via web browsers on computers or smartphones.
- Users will possess valid institutional email credentials (@lnmiit.ac.in).
- Wellness staff and counselors are qualified institutional psychological professionals.
- Student counseling inquiries and habit logs require strict privacy and role-restricted confidentiality.
- Students will log self-care activities voluntarily and honestly for personal habit tracking.
- The system will initially be developed as a responsive web application.
- The system will use a centralized relational database (e.g., MySQL/PostgreSQL).

9. Initial Constraints

- Limited development time aligned with academic semester deadlines.
- Student development team with limited personnel resources.
- Limited access to sensitive live institutional psychiatric data during development (simulated seed data utilized).
- Strict ethical and legal boundaries precluding medical claims or automated psychiatric diagnosis.
- Requirement changes during incremental development cycles.
- Need to protect user identity and confidential consultation requests.
- Limited field testing with real student cohorts prior to institutional administrative clearance.

10. Initial Success Criteria

Student Wellness Portal will be considered successful if the team can demonstrate that:

- different user roles can access appropriate functionality;
- a student can discover wellness services and submit a confidential support request;
- an authorized counselor can triage requests and manage consultation slots;
- a counseling appointment can be booked without scheduling conflicts;
- students can log daily self-care habits and view visual trend graphs;
- psychoeducational resources can be published by staff and accessed by students;
- automated notifications are delivered upon status transitions and appointment bookings;
- administrative dashboards accurately display aggregate wellness statistics;
- important system functions are tested and defects are documented and resolved;
- the project is maintained in a controlled Git/GitHub repository;
- the final system is accompanied by appropriate software engineering documentation.

11. User Classes

11.1 Student

Students are the primary participants and users of the portal. They shall be able to:

- authenticate securely using institutional credentials;
- view and search the catalog of available wellness and counseling services;
- submit structured, confidential support requests with urgency categorization;
- book, reschedule, and cancel counseling consultation appointments;
- view real-time request statuses and consultation schedules;
- log daily self-care habits (sleep, hydration, meditation, workouts) and mood scores;
- view visual habit streak analytics and 7-day/30-day wellbeing trend graphs;
- browse, read, and bookmark psychoeducational awareness articles;

- receive automated in-app notifications and appointment reminders.

11.2 Wellness Staff / Counselor

The counselor is responsible for clinical triage, consultations, and psychoeducation. The counselor shall be able to:

- authenticate with elevated staff privileges;
- access a dedicated triage queue displaying assigned student support cases;
- review submitted student requests and assign clinical priority;
- publish, edit, and manage weekly recurring appointment availability slots;
- record confidential clinical notes, session summaries, and follow-up recommendations;
- escalate critical emergency cases to campus medical emergency units;
- publish and maintain psychoeducational articles in the resource library;
- record session attendance (Attended, No-Show, Rescheduled, Cancelled).

11.3 Health Center / Medical Referral Specialist

The health center medical specialist shall be able to:

- review escalated high-risk emergency support cases;
- provide clinical intake feedback and acknowledge medical emergency referrals;
- coordinate immediate physical health center intervention when warranted;
- maintain emergency audit records in compliance with health safety policies.

11.4 Administrator

The administrator shall be able to:

- manage user accounts and assign/modify role permissions (Student, Staff, Admin);
- configure wellness services, categories, and operational parameters;
- broadcast urgent campus-wide wellness advisories and announcements;
- monitor system security, active sessions, and immutable audit logs;
- access administrative operational summaries and performance metrics.

11.5 Institute Management

Management users shall have access to appropriate aggregated, anonymized reports and dashboards for monitoring campus wellness trends, service demand, and resolution metrics.

12. System Context

The system context diagram illustrates the boundary of the Student Wellness Portal and its interactions with external entities (Student, Wellness Staff / Counselor, Campus Health Center & Medical Referrals, Campus Administrator, Institute Management, and Institutional Email Verification & SMTP Server).

Figure 12.1: System Context Diagram (Level-0 Context DFD) for Student Wellness Portal

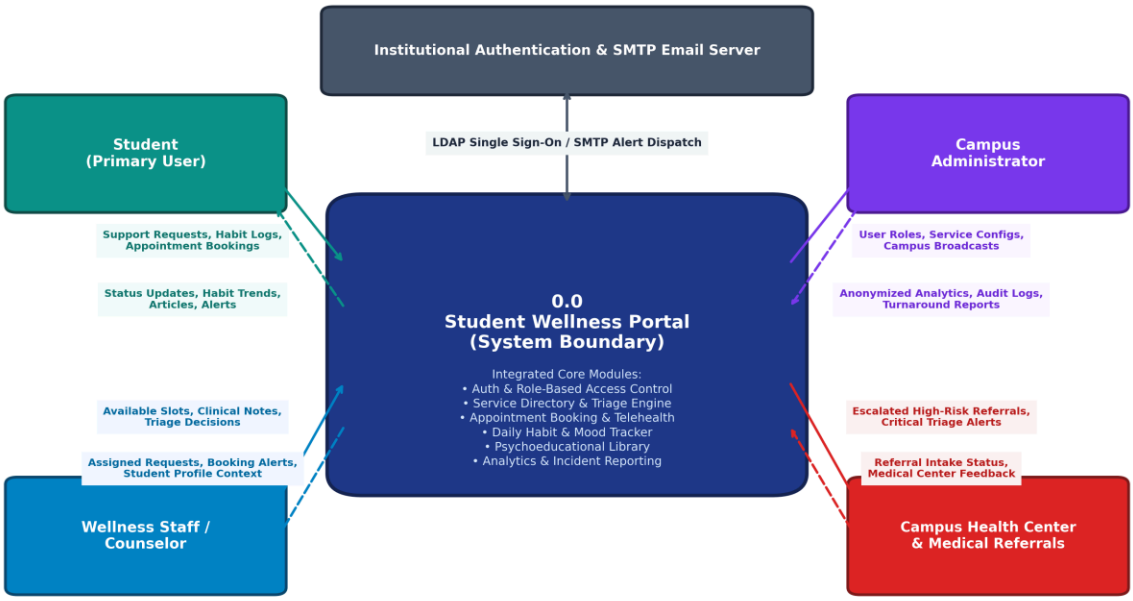


Figure 12.1: System Context Diagram (Level-0 Context DFD) for Student Wellness Portal

13. Functional Requirements

This section specifies exactly 70 Functional Requirements (FR-001 to FR-070) categorized across all primary functional modules:

13.1 Authentication and Account Management

FR-001 — User Registration: The system shall allow students and institutional staff to register using verified institutional email addresses (@lnmiit.ac.in).

FR-002 — User Login & Authentication: The system shall allow registered users to authenticate securely using their institutional email and password credentials.

FR-003 — Authentication Failure Handling: The system shall notify the user of failed login attempts without revealing whether the email or password was incorrect.

FR-004 — Session Management & Logout: The system shall allow authenticated users to terminate their active session and automatically expire sessions after 30 minutes of inactivity.

FR-005 — Role-Based Access Control (RBAC): The system shall enforce role-specific access permissions strictly segregating Student, Counselor/Staff, and Administrator functions.

FR-006 — Administrative Role Assignment: The system shall allow authorized administrators to assign, modify, and revoke user roles.

FR-007 — Password Reset & Recovery: The system shall provide a secure password reset workflow via time-limited, single-use email verification tokens.

FR-008 — User Account Deactivation: The system shall allow administrators to suspend or deactivate user accounts while preserving historical audit logs.

13.2 Student Wellness Profile & Onboarding

FR-009 — Initial Profile Setup: The system shall prompt first-time student users to configure their wellness profile, emergency contact details, and confidentiality preferences.

FR-010 — Profile Modification: The system shall allow students to update their personal contact information, academic branch, and notification preferences at any time.

FR-011 — Anonymity Settings: The system shall allow students to toggle alias-based anonymous display for public peer discussions while preserving identity for clinical appointments.

FR-012 — Confidential Emergency Contact: The system shall store emergency contact details in an encrypted format accessible only during designated clinical emergency escalations.

FR-013 — Wellness Goal Setting: The system shall allow students to define personal self-care goals (target sleep hours, weekly exercise frequency, daily meditation minutes, water intake).

FR-014 — Initial Wellness Questionnaire: The system shall provide a non-diagnostic self-guided wellness onboarding questionnaire to suggest relevant campus services.

FR-015 — Profile Data Export: The system shall allow students to export their recorded wellness logs and history in a standardized PDF/JSON format.

13.3 Wellness Service Catalog & Discovery

FR-016 — Service Catalog Display: The system shall display all active institutional wellness services categorized by domain (Mental Health, Academic Stress, Nutrition & Fitness, Peer Support).

FR-017 — Service Detail View: The system shall display comprehensive details for each service, including service description, counselor bios, operating hours, venue, and confidentiality policies.

FR-018 — Service Keyword Search: The system shall allow users to search for services using keywords matching service titles, descriptions, and counseling tags.

FR-019 — Multi-Criteria Service Filtering: The system shall allow users to filter services by category, delivery mode (In-person, Online Telehealth), and available counselor specialization.

FR-020 — Service Availability Status: The system shall indicate the real-time operational status (Active, Temporarily Full, On Leave) for each wellness service.

FR-021 — Service Creation & Publishing: The system shall allow administrators and authorized staff to create and publish new wellness service entries.

FR-022 — Service Modification: The system shall allow administrators to update service descriptions, capacity limits, and assigned staff members.

FR-023 — Service Archival: The system shall allow administrators to archive obsolete services while preventing disruption to existing scheduled appointments.

13.4 Service Request & Triage Lifecycle

FR-024 — Multi-Step Request Submission: The system shall allow students to submit structured support requests detailing concern category, urgency rating, preferred mode, and background description.

FR-025 — Request Urgency Classification: The system shall require students to classify request urgency (Routine, Priority, Urgent) with guided situational examples.

FR-026 — Urgent Request Flagging: The system shall immediately flag urgent requests on the counselor triage queue and trigger priority alerts to the on-duty counselor.

FR-027 — Request Active Limit Validation: The system shall prevent a student from submitting a new service request if they have two or more pending active requests.

FR-028 — Request Status Tracking: The system shall provide students with a real-time status tracker displaying current stage (Pending Review, Triaged, Scheduled, In Progress, Completed, Closed).

FR-029 — Request Assignment: The system shall allow wellness coordinators to assign incoming service requests to specific qualified counselors.

FR-030 — Student Request Cancellation: The system shall allow a student to withdraw or cancel a pending support request with a mandatory reason prompt.

FR-031 — Counselor Reassignment: The system shall allow staff to reassign cases to alternate counselors in the event of leave or specialized clinical needs.

FR-032 — Request Timeline Audit: The system shall record timestamped logs for all status transitions and staff assignments associated with every service request.

13.5 Counseling Appointment Scheduling & Booking

FR-033 — Slot-Based Availability Calendar: The system shall display real-time calendar slots published by counselors for student booking.

FR-034 — Appointment Booking: The system shall allow eligible students with triaged requests to select and confirm an available counseling appointment slot.

FR-035 — Conflict & Double-Booking Prevention: The system shall ensure that booked time slots are locked atomically to prevent simultaneous double-booking of counselors.

FR-036 — Advance Notice Enforcement: The system shall enforce a minimum 24-hour advance booking rule for routine non-emergency appointments.

FR-037 — Telehealth Link Generation: The system shall automatically generate and attach a secure, unique video consultation link for confirmed virtual appointments.

FR-038 — Appointment Rescheduling: The system shall allow students and counselors to reschedule confirmed appointments at least 6 hours before the scheduled time.

FR-039 — Appointment Cancellation: The system shall allow participants to cancel appointments at least 4 hours in advance, automatically reopening the slot for other students.

FR-040 — Automated Appointment Reminders: The system shall dispatch automated reminder notifications to both student and counselor 24 hours and 1 hour prior to scheduled appointments.

FR-041 — Attendance & Session Check-In: The system shall allow counselors to record attendance status (Attended, No-Show, Rescheduled, Cancelled) upon session conclusion.

13.6 Counselor Workflow, Case Notes & Clinical Workflow

FR-042 — Counselor Triage Queue: The system shall provide counselors with a prioritized triage dashboard highlighting pending cases, urgency badges, and wait times.

FR-043 — Confidential Clinical Notes Logging: The system shall allow assigned counselors to record confidential clinical consultation notes, observations, and recommendations.

FR-044 — Clinical Notes Access Restriction: The system shall restrict visibility of clinical notes exclusively to the authoring counselor and authorized medical center leads.

FR-045 — Counselor Schedule & Slot Management: The system shall allow counselors to configure weekly recurring available slots, block out leave periods, and set consultation limits.

FR-046 — Emergency Escalation & Medical Referral: The system shall allow counselors to initiate an emergency referral protocol directing high-risk cases to campus medical emergency services.

FR-047 — Case Closure & Summary Report: The system shall require counselors to submit closing remarks and outcome categorization before marking a case as Completed.

FR-048 — Follow-Up Appointment Scheduling: The system shall allow counselors to book direct follow-up consultation slots for recurring student therapy plans.

13.7 Daily Habit & Self-Care Activity Tracking

FR-049 — Daily Activity Logging: The system shall allow students to record daily self-care metrics including sleep duration, water intake, physical activity, and mindfulness meditation.

FR-050 — Daily Mood Check-In: The system shall allow students to submit a daily mood score (1–5 scale) accompanied by optional reflection journal tags.

FR-051 — Daily Habit Streak Calculation: The system shall compute and update consecutive daily logging streaks for each tracked habit category.

FR-052 — Visual Habit Trend Analytics: The system shall render 7-day and 30-day interactive graphical trend charts illustrating student wellness activity and sleep patterns.

FR-053 — Historical Habit Log Review: The system shall allow students to browse and review their past daily logs through an interactive calendar view.

FR-054 — Habit Backlog Constraint: The system shall permit students to log or edit habit entries up to a maximum of 48 hours in the past to maintain data integrity.

FR-055 — Encouragement & Milestone Badges: The system shall award non-competitive positive encouragement badges upon reaching habit consistency milestones (e.g., 7-day streak).

FR-056 — Habit Data Privacy Guarantee: The system shall ensure individual habit logs remain completely private to the student and are never exposed to counselors without explicit consent.

13.8 Psychoeducational Resource Library & Content Management

FR-057 — Resource Catalog Listing: The system shall display a curated library of psychoeducational articles, anxiety management guides, and mindfulness audio guides.

FR-058 — Resource Categorization & Tags: The system shall categorize resources under thematic tags (Stress Management, Sleep Hygiene, Exam Coping, Nutrition, Mindfulness).

FR-059 — Resource Reader & Content Viewer: The system shall provide an accessible, responsive reading interface with estimated read times and author credentials.

FR-060 — Resource Publishing & Editing: The system shall allow authorized counselors and administrators to draft, edit, publish, and update resource articles.

FR-061 — Student Resource Bookmarking: The system shall allow students to bookmark favourite articles and self-help guides for rapid offline/dashboard access.

FR-062 — Resource Media Attachment: The system shall support rich-text articles embedded with approved downloadable PDF infographics and guided audio exercises.

13.9 Automated Notification & Alert System

FR-063 — In-App Notification Center: The system shall maintain an in-app notification center displaying real-time alert badges and chronologically ordered messages.

FR-064 — Multi-Trigger Event Notifications: The system shall dispatch instant notifications upon request submission, status changes, slot confirmations, and counselor notes updates.

FR-065 — Notification Read State Management: The system shall allow users to mark notifications as read individually or collectively mark all as read.

FR-066 — Campus-Wide Wellness Broadcasts: The system shall allow administrators to broadcast urgent institutional wellness advisories, workshop alerts, and helpline updates.

13.10 Administrative Analytics, Reporting & System Config

FR-067 — Anonymized Institutional Metrics Dashboard: The system shall render administrative analytics summarizing aggregate service demand, popular categories, and monthly case volumes.

FR-068 — Counselor Workload & Turnaround Reports: The system shall generate anonymized reports on average case resolution turnaround times and counselor consultation load distribution.

FR-069 — System Audit Log Tracking: The system shall record an immutable audit log of administrative actions, role modifications, and system configuration adjustments.

FR-070 — Compliance & Aggregate Data Export: The system shall allow administrators to export aggregate institutional wellness statistics in CSV/PDF format for annual accreditation reporting.

14. Business Rules

ID	Business Rule
BR-001	Only authenticated users with active institutional credentials (@lnmiit.ac.in) can access portal features.

BR-002	A student cannot have more than two active open support requests simultaneously.
BR-003	Routine non-emergency counseling appointments must be booked at least 24 hours in advance.
BR-004	Counselors cannot be double-booked for overlapping time slots; slot selection is atomic.
BR-005	Clinical consultation notes and remarks are confidential and only accessible to assigned counselors and medical center leads.
BR-006	Self-care and habit tracking logs are strictly private to the student and are never exposed to counselors without explicit student consent.
BR-007	Urgent/Emergency priority requests must automatically trigger emergency escalation protocols and immediate high-priority counselor alerts.
BR-008	Only authorized wellness staff and administrators can publish, modify, or archive wellness awareness resources.
BR-009	Appointment cancellations must be executed at least 4 hours prior to the scheduled start time to release the slot for other students.
BR-010	Deleted or archived user accounts retain anonymized historical records for institutional accreditation and audit compliance.
BR-011	Telehealth virtual consultation meeting URLs are generated dynamically only upon confirmation of an appointment.
BR-012	Daily habit entries can only be backlogged within a 48-hour historical window to maintain clinical tracking integrity.
BR-013	Administrative reports must display aggregate, strictly anonymized metrics without exposing student identifying information.
BR-014	Only System Administrators are authorized to modify user roles, assign elevated privileges, and manage system parameters.
BR-015	Completed consultations require mandatory counselor closing remarks and outcome categorization prior to case archiving.

15. Non-Functional Requirements

15.1 Performance

NFR-001: The system shall provide page load and navigation response times within 1.5 seconds under normal operational load.

NFR-002: Search, filtering, and catalog queries shall return results within 500 milliseconds for active datasets.

NFR-003: The system shall support at least 500 concurrent active users without degradation of transactional integrity.

15.2 Security

NFR-004: All user authentication data and API communications shall be encrypted in transit using TLS 1.3.

NFR-005: Sensitive user passwords shall be hashed using bcrypt with a work factor of 12 prior to database storage.

NFR-006: Role-Based Access Control (RBAC) shall be strictly enforced on both frontend routes and backend REST endpoints.

NFR-007: All administrative and clinical triage operations shall generate tamper-evident audit logs.

15.3 Usability

NFR-008: The user interface shall provide clean, consistent, and empathetic navigation adhering to WCAG 2.1 Level AA accessibility guidelines.

NFR-009: User-facing error messages shall be friendly, constructive, and avoid exposing internal system stack traces.

NFR-010: Support request submission and habit logging workflows shall require no more than 3 user clicks from the dashboard.

15.4 Reliability

NFR-011: Support request submissions and appointment booking transactions shall exhibit 99.9% data persistence reliability.

NFR-012: The database layer shall implement automatic rollback mechanisms to prevent orphaned records during booking conflicts.

15.5 Availability

NFR-013: The system shall maintain an operational uptime availability of 99.5% during academic semester cycles, excluding scheduled maintenance.

15.6 Maintainability

NFR-014: The codebase shall follow a modular, decoupled architecture (Frontend, Backend API, Database) with clear separation of concerns.

NFR-015: All source code and database migration scripts shall be maintained under Git version control.

NFR-016: Comprehensive API documentation and software requirements specifications shall be updated in synchronization with codebase changes.

15.7 Scalability

NFR-017: The database and backend architecture shall support horizontal scaling to accommodate growing student cohorts and longitudinal habit logs.

15.8 Compatibility

NFR-018: The web application shall render responsively across major modern browsers (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari) on desktop, tablet, and mobile devices.

15.9 Privacy

NFR-019: Student mental health notes, request descriptions, and habit histories shall be accessible only to authorized stakeholders and masked in management reports.

15.10 Auditability

NFR-020: High-privilege administrative actions, role assignments, and case escalations shall be permanently logged with timestamps and actor IDs.

16. Data Requirements

The initial logical information requirements encompass the following primary data entities and attributes:

Entity	Attributes & Descriptions
User	UserID, FullName, Email (@lnmiit.ac.in), PasswordHash, Role (Student | Counselor | HealthCenter | Admin | Management), RollNumber, Department, ContactNumber, AccountStatus (Active | Suspended), CreatedAt
WellnessProfile	ProfileID, UserID (FK), EmergencyContactName, EmergencyContactPhone, EmergencyContactRelation, PrimaryConcerns, PreferredCounselingMode, AnonymityAlias, OnboardingCompleted
WellnessService	ServiceID, Title, Category (Mental Health | Academic Stress | Nutrition | Fitness), Description, AssignedCounselorID (FK), Mode (In-person | Telehealth), Venue, DailyCapacity, Status (Active | Inactive)
ServiceRequest	RequestID, StudentID (FK), ServiceID (FK), Category, UrgencyLevel (Routine | Priority | Urgent), Description, PreferredSchedule, Status (Pending Review | Triaged | Scheduled | In Progress | Completed | Cancelled | Escalated), AssignedCounselorID (FK), CreatedAt, UpdatedAt
Appointment	AppointmentID, RequestID (FK), StudentID (FK), CounselorID (FK), AppointmentDate,

	StartTime, EndTime, Mode (In-person | Telehealth), MeetingLink, Venue, Status (Confirmed | Rescheduled | Completed | Cancelled | No-Show), CreatedAt
ClinicalNote	NoteID, AppointmentID (FK), RequestID (FK), CounselorID (FK), StudentID (FK), SessionSummary, ClinicalObservations, ActionPlan, ConfidentialityLevel, RecordedAt
HabitLog	LogID, StudentID (FK), LogDate, SleepHours, WaterIntakeLiters, MeditationMinutes, WorkoutMinutes, MoodScore (1-5), JournalTag, StreakCount, CreatedAt
ResourceArticle	ArticleID, Title, Category, Summary, ContentBody, AuthorStaffID (FK), Tags, ReadTimeMinutes, ThumbnailURL, Status (Draft | Published | Archived), PublishedDate
Notification	NotificationID, RecipientID (FK), Type (RequestUpdate | BookingReminder | EscalationAlert | Broadcast), Title, Message, ActionURL, IsRead, CreatedAt
AdminAuditLog	AuditID, ActorUserID (FK), ActionType, TargetEntity, EntityID, IPAddress, Timestamp, Details

17. Service Request & Consultation Lifecycle

A student support case passes through the following canonical states: Draft / Submission ➔ Pending Review (Triage Queue) ➔ Triaged / Assigned to Counselor ├── [Routine / Priority] ➔ Slot Booking ➔ Scheduled ➔ In Progress ➔ Completed (Notes Recorded) └── [Critical Emergency] ➔ Emergency Escalation ➔ Campus Health Center Referral ➔ Medical Follow-up (Alternative: Cancelled / Withdrawn by Student).

```
Draft / Submission
 ↓
Pending Review (Triage Queue)
 ↓
Triaged / Assigned to Counselor
 ├── [Routine / Priority] ➔ Slot Booking ➔ Scheduled ➔ In Progress ➔
Completed (Notes Recorded)
 └── [Critical Emergency] ➔ Emergency Escalation ➔ Campus Health Center
Referral ➔
Medical Follow-up
 (Alternative: Cancelled / Withdrawn by Student)
```

18. Major Use Cases

ID	Use Case	Primary Actor

UC-001	User Authentication & Session Logout	All Users (Student, Counselor, Admin)
UC-002	Manage User Accounts & Assign Roles	Campus Administrator
UC-003	Configure Student Wellness Profile & Goals	Student
UC-004	Browse & Search Wellness Service Directory	Student
UC-005	Submit Confidential Wellness Support Request	Student
UC-006	Track Support Request Status Lifecycle	Student
UC-007	Book Counseling Consultation Appointment	Student
UC-008	Reschedule or Cancel Appointment	Student / Counselor
UC-009	Manage Counselor Availability Slots	Wellness Staff / Counselor
UC-010	Conduct Clinical Triage & Case Assignment	Wellness Staff / Counselor
UC-011	Escalate Emergency Case to Health Center	Wellness Staff / Counselor
UC-012	Log Daily Self-Care Habits & Mood Check-In	Student
UC-013	View Visual Habit Streaks & Trend Graphs	Student
UC-014	Publish & Manage Psychoeducational Articles	Wellness Staff / Counselor
UC-015	Browse & Read Awareness Resources	Student
UC-016	Bookmark Wellness Resource Articles	Student
UC-017	Dispatch & View In-App Notifications	System / All Users
UC-018	Broadcast Campus-Wide Wellness Announcements	Campus Administrator
UC-019	Generate Aggregate Institutional Analytics	Campus Administrator / Institute Management
UC-020	Inspect System Audit & Security Logs	Campus Administrator

19. Detailed Use Cases

Detailed Use Case 1: UC-005 — Submit Confidential Wellness Support Request
Primary Actor: Student

Preconditions: Student is authenticated; student has fewer than 2 active open support requests; selected service is active.

Main Flow:

- 1. Student navigates to the Wellness Services Catalog.
- 2. Student browses/searches for a service and selects 'Request Support'.
- 3. System presents the multi-step confidential request form.
- 4. Student selects urgency rating (Routine, Priority, Urgent).
- 5. Student selects preferred consultation mode (In-person or Online Telehealth).
- 6. Student enters background description and preferred time availability.
- 7. Student reviews confidentiality notice and submits the request.
- 8. System validates active request limits and inputs.
- 9. System persists the request with status PENDING REVIEW.
- 10. System generates a confirmation notification for the student and alerts the counseling triage queue.

Alternative / Exception Flows: If student already has 2 pending requests, system blocks submission; if urgency is 'Urgent', system displays 24/7 crisis numbers and routes ticket to high-priority triage.

Postcondition: A valid service request record is persisted with status PENDING REVIEW.

Detailed Use Case 2: UC-007 — Book Counseling Consultation Appointment

Primary Actor: Student

Preconditions: Student has an approved/triaged support request; assigned counselor has open availability slots.

Main Flow:

- 1. Student opens their triaged support request and clicks 'Schedule Appointment'.
- 2. System retrieves and renders the counselor's calendar with available slots.
- 3. Student selects a desired date and 45-minute time slot.
- 4. System verifies slot availability and enforces the 24-hour advance booking rule.
- 5. Student confirms appointment mode (In-Person or Telehealth) and submits booking.
- 6. System atomically locks the slot and creates an Appointment record with status CONFIRMED.
- 7. If Telehealth mode is selected, system generates a secure virtual consultation URL.
- 8. System updates the associated ServiceRequest status to SCHEDULED.
- 9. System sends booking confirmation notifications to both student and counselor.

Alternative / Exception Flows: If slot is taken concurrently, system aborts lock and refreshes open slots; if slot is within 24 hours for routine request, booking is restricted.

Postcondition: An appointment record with status CONFIRMED is created, slot is locked, and notifications sent.

Detailed Use Case 3: UC-010 — Conduct Clinical Triage & Case Assignment

Primary Actor: Wellness Staff / Counselor

Preconditions: Counselor is authenticated with elevated privileges; student support requests exist in PENDING REVIEW status.

Main Flow:

- 1. Counselor accesses the Staff Triage Dashboard.
- 2. System presents incoming requests sorted by urgency and submission timestamp.

- 3. Counselor reviews student background concerns, urgency categorization, and history.
- 4. Counselor assigns clinical priority level and allocates case to assigned counselor.
- 5. Counselor submits triage evaluation.
- 6. System updates ServiceRequest status to TRIAGED.
- 7. System dispatches notification to student unlocking appointment slot booking calendar.
- 8. System logs triage action in the tamper-evident audit trail.

Alternative / Exception Flows: If acute self-harm risk or psychiatric emergency is identified, counselor triggers emergency escalation (UC-011), updating status to ESCALATED and alerting Campus Health Center.

Postcondition: ServiceRequest status is updated to TRIAGED (or ESCALATED), student is notified, and audit log is recorded.

20. User Stories

US-001: As a student, I want to securely log in using my institutional email, so that my personal wellness information remains strictly confidential.

US-002: As a student, I want to browse categorized campus wellness services, so that I can discover available psychological, nutritional, and academic support.

US-003: As a student, I want to submit a confidential support request with urgency ratings, so that counselors can prioritize my situation appropriately.

US-004: As a student, I want to book an available consultation slot on a calendar, so that I can meet a counselor without scheduling conflicts.

US-005: As a student, I want to join online counseling sessions via a secure Telehealth link, so that I can receive guidance remotely from my hostel or home.

US-006: As a student, I want to track daily habits (sleep, water, meditation, workouts) and mood, so that I can monitor my personal self-care consistency.

US-007: As a student, I want to view 7-day and 30-day habit trend graphs, so that I can visually assess my wellbeing progress over time.

US-008: As a student, I want to read and bookmark psychoeducational articles, so that I can learn evidence-based anxiety management techniques.

US-009: As a student, I want to receive automated appointment reminders, so that I do not miss scheduled counseling sessions.

US-010: As a counselor, I want to view a prioritized triage queue of incoming student cases, so that urgent cases receive immediate clinical attention.

US-011: As a counselor, I want to manage my weekly availability slots, so that students can only book during my designated consultation hours.

US-012: As a counselor, I want to log confidential session notes and action plans, so that I can maintain continuity of care for recurring consultations.

US-013: As a counselor, I want to escalate high-risk cases to the campus health center, so that critical medical interventions can occur immediately.

US-014: As an administrator, I want to manage user accounts and assign roles, so that access to sensitive clinical functions is strictly controlled.

US-015: As an administrator / management user, I want to view aggregated, anonymized wellness analytics, so that I can evaluate campus mental health trends and allocate resources effectively.

US-016: As a campus health center specialist, I want to receive immediate alerts for escalated high-risk cases, so that urgent physical medical or psychiatric interventions can be coordinated without delay.

US-017: As a student or counselor, I want to reschedule or cancel an appointment with advance notice, so that consultation slots can be freed up for others.

US-018: As a counselor, I want to author and publish self-help psychoeducational articles, so that students can access reliable mental health resources.

US-019: As an administrator, I want to broadcast urgent campus-wide announcements and manage audit logs, so that students receive critical wellness advisories and governance is maintained.

21. Acceptance Criteria Example

US-003: Submit Confidential Wellness Support Request

ID	Acceptance Criterion
AC-01	Student must be authenticated with valid institutional credentials (@lnmiit.ac.in).
AC-02	Student must have fewer than 2 active open support requests.
AC-03	Selected wellness service must be in Active status.
AC-04	Student must provide concern category, urgency rating (Routine, Priority, Urgent), and description.
AC-05	Urgent rating selection must display immediate campus 24/7 crisis helpline details.
AC-06	System must persist the request with status PENDING REVIEW.
AC-07	Student must receive an in-app confirmation notification with a unique Request ID.
AC-08	Request must appear immediately on the counselor triage queue with appropriate urgency badges.

US-004: Book Counseling Consultation Appointment

ID	Acceptance Criterion
AC-09	Student must possess an approved or triaged support request.
AC-10	System must display only currently available, unbooked counselor slots.
AC-11	Routine appointments must enforce a minimum 24-hour advance booking rule.

AC-12	Selected slot must be locked atomically during booking to prevent double-booking race conditions.
AC-13	Virtual appointments must generate and attach a secure, unique Telehealth URL.
AC-14	System must create an Appointment record with status CONFIRMED.
AC-15	Automated booking confirmations must be dispatched to both student and counselor.

US-006: Log Daily Self-Care Habits & Mood

ID	Acceptance Criterion
AC-16	Student must be able to log sleep hours, water intake, meditation minutes, and workouts.
AC-17	Student must be able to select a mood score from 1 to 5 with optional journal reflection tags.
AC-18	System must validate that numerical inputs fall within physiological limits (e.g., sleep 0-24 hrs).
AC-19	System must update consecutive daily habit streak counts automatically.
AC-20	System must recalculate and update 7-day and 30-day graphical habit trend visualizations.
AC-21	Habit records must remain strictly private to the student and inaccessible to other users.

22. Requirements Prioritization (MoSCoW)

Category	Requirements / Features Included
Must Have (Core Critical)	• Institutional user authentication & RBAC (FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-008) • Student Profile & Privacy Controls (FR-009, FR-010, FR-011, FR-012) • Wellness service catalog & search (FR-016, FR-017, FR-018, FR-019, FR-020) • Support request submission & urgency triage queue (FR-024, FR-025, FR-027, FR-028, FR-029, FR-042) • Emergency health center escalation protocol (FR-026, FR-046, BR-007) • Slot-based appointment booking & double-booking prevention (FR-033, FR-034, FR-035, FR-036, FR-045) • Confidential clinical consultation notes logging (FR-043, FR-044, FR-047)

	• Automated in-app notifications for status transitions & reminders (FR-063, FR-064, FR-065)
Should Have (High Priority)	• Password reset & data export (FR-007, FR-015) • Daily self-care habit logging (sleep, hydration, meditation, workouts) & mood check-in (FR-049, FR-050, FR-054, FR-056) • 7-day & 30-day visual habit trend analytics & streaks (FR-051, FR-052, FR-053) • Psychoeducational resource library & counselor CMS authoring (FR-057, FR-058, FR-059, FR-060, FR-062) • Telehealth virtual consultation link generation (FR-037) • Appointment rescheduling & cancellation workflows (FR-030, FR-038, FR-039, FR-041) • Anonymized institutional aggregate wellness analytics dashboard (FR-067, FR-068, FR-070) • Campus-wide admin broadcast advisories & immutable audit logs (FR-066, FR-069)
Could Have (Desirable)	• Personal wellness goal setting & habit milestone badges (FR-013, FR-055) • Self-guided onboarding wellness questionnaire (FR-014) • Dynamic service creation & catalog management (FR-021, FR-022, FR-023) • Counselor case reassignment & follow-up scheduling (FR-031, FR-048) • Student article bookmarking & favorites collection (FR-061) • Automated pre-session reminder notifications (FR-040)
Won't Have (Out of Scope)	• Automated psychiatric medical diagnosis or prescription dispensing • Continuous wearable sensor / smartwatch hardware synchronization • Paid commercial payment gateway integration • University academic grade synchronization or hospital billing EHR systems • Native custom video streaming server infrastructure (external URLs utilized)

	• LDAP / SAML Single Sign-On integration (post-v1.0 enhancement)

23. Requirements Traceability Matrix (RTM)

Functional Requirements	User Story	Use Case	Test
FR-001, FR-002, FR-003, FR-004, FR-007	US-001	UC-001	Unit & Integration Test: Auth & Token Reset
FR-005, FR-006, FR-008	US-019	UC-002	Security Test: RBAC Middleware & Accounts
FR-009, FR-010, FR-011, FR-012, FR-013, FR-015	US-003	UC-003	Functional Test: Profile & Privacy Controls
FR-014, FR-016, FR-018, FR-019, FR-020	US-002	UC-004	UI Test: Service Search & Filters
FR-017, FR-021, FR-022, FR-023	US-002	UC-004	Functional Test: Service Catalog Management
FR-024, FR-025, FR-026, FR-027	US-004	UC-005	Functional Test: Request Form & Urgency
FR-028, FR-030, FR-032	US-004	UC-006	Workflow Test: Request Lifecycle & Audit
FR-033, FR-034, FR-035, FR-036	US-005	UC-007	Concurrency Test: Slot Booking Lock
FR-037, FR-038, FR-039	US-006, US-017	UC-008	Integration Test: Telehealth & Reschedule
FR-029, FR-031, FR-041, FR-042	US-012	UC-010	Functional Test: Clinical Triage & Assignment
FR-043, FR-044, FR-047	US-014	UC-010	Security & Privacy Test: Clinical Session Notes
FR-045, FR-048	US-013	UC-009	Functional Test: Counselor Slot Management
FR-046	US-015, US-016	UC-011	Integration Test: Emergency Health Escalation

FR-049, FR-050, FR-054, FR-056	US-007	UC-012	Functional Test: Daily Habit & Mood Logging
FR-051, FR-052, FR-053, FR-055	US-008	UC-013	Analytics Test: Habit Streaks & Trends
FR-057, FR-058, FR-059	US-009	UC-015	UI Test: Psychoeducational Resource Reader
FR-060, FR-062	US-018	UC-014	CMS Test: Psychoeducational Publishing
FR-061	US-010	UC-016	Functional Test: Student Bookmarking
FR-040, FR-063, FR-064, FR-065	US-011	UC-017	Integration Test: In-App Notifications
FR-066	US-019	UC-018	Functional Test: Campus-Wide Admin Broadcasts
FR-067, FR-068, FR-070	US-015	UC-019	Reporting Test: Aggregate Analytics
FR-069	US-019	UC-020	Security Test: Audit Log Integrity

Note: The TBD entries will be executed and verified during the subsequent software testing and verification phase.

24. Requirements Validation

Quality Attribute	Validation Review & Criteria Assessment
Correctness	Requirements accurately capture genuine student wellness, counseling triage, habit tracking, and administrative governance needs.
Unambiguity	Every functional requirement is articulated using definitive modal phrasing ('The system shall...') with single interpretations.
Completeness	All functional domains, edge cases, error conditions, and data entities are comprehensively specified across 70 FRs.
Consistency	No contradictions exist between functional specifications, business rules, security policies, and non-functional constraints.
Verifiability	Each requirement contains concrete, testable acceptance criteria suitable for automated or manual QA test verification.
Feasibility	All planned features are technically achievable within the semester timeline

	utilizing modern MERN/web stack technologies.
Traceability	Every requirement links bi-directionally to User Stories, Use Cases, and downstream testing items via the RTM.
Prioritization	Features are classified transparently via MoSCoW methodology to guide agile sprint planning and iterative releases.
Modifiability	Requirements are hierarchically numbered and modularly structured so changes can occur with minimal ripple effects.

25. Example of Requirement Quality

Poor Requirement (Ambiguous & Untestable):

'The system should be fast, very secure, and make booking counseling appointments easy for students.'

Deficiency Analysis: 'Fast', 'very secure', and 'easy' are subjective, non-quantifiable adjectives that cannot be objectively verified or tested.

Better Requirements (Precise, Specific & Testable):

- 'The system shall provide appointment slot booking and conflict-verification responses within 500 milliseconds under a concurrent load of 500 active users.' (Specific, Quantifiable Performance)
- 'The system shall enforce TLS 1.3 encryption for all data in transit and hash user passwords using bcrypt with a work factor of 12.' (Definitive Security Specification)
- 'The system shall allow authenticated students to complete an appointment booking in no more than 4 form interactions from the service catalog view.' (Measurable Usability Metric)

26. Requirement Change Management

Because the Student Wellness Portal follows an Agile + Incremental development framework, requirement modifications are handled via a structured change control workflow:

Change Request ➔ Impact & Feasibility Analysis ➔ Requirement Specification Update ➔ Backlog Reprioritization ➔ Sprint Planning ➔ Design & Architecture Adjustment ➔ Implementation & Unit Testing ➔ Traceability Matrix Sync ➔ Baseline Release

Example Change Scenario: If institutional stakeholders request QR-code attendance check-in for in-person workshops, an impact analysis is conducted to evaluate database schema changes, UI adjustments, and security implications prior to backlog inclusion.

27. Future Requirements

Potential enhancements identified for post-v1.0 releases include:

- Automated waitlist management for fully booked counseling slots with auto-notification on cancellations;
- Calendar synchronization integration (Google Calendar, Outlook iCal export/sync);
- AI-driven personalized self-care psychoeducational reading recommendations based on habit trends;
- Dedicated cross-platform mobile application (React Native / Flutter);

- Single Sign-On (SSO) integration with LNMIIT Institutional Microsoft / Google Workspace LDAP;
- Interactive peer support discussion forums with sentiment moderation and automated distress detection;
- Automated clinical outcome tracking surveys (e.g., standardized PHQ-9 / GAD-7 non-diagnostic wellness screeners);
- Integration with campus physical fitness centers for gym/sports activity validation.

28. SRS Baseline and Versioning

- Initial Baseline: SRS v1.0 (Lab Assignment 2 Release)
- Version Control Information: Baseline SRS v1.0 establishes the complete requirements specification for the Student Wellness Portal, specifying 70 functional requirements, 20 non-functional requirements, 15 business rules, data schemas, use case catalogs, and traceability matrices to govern subsequent UML design and implementation phases.

29. Individual Contribution

Student Name & Roll No.	Individual Contribution
Mehul Baid (24UCS003)	Formulated functional requirements FR-001 to FR-035, developed the Level-0 System Context DFD diagram, authored User Classes, and structured the baseline project specification.
Alakh Agarwal (24UCS001)	Formulated functional requirements FR-036 to FR-070, defined the 15 Business Rules, and authored the Non-Functional Requirements (NFR-001 to NFR-020).
Yug Nahar (24UCS004)	Formulated Data Requirements (10 core entities & schemas), developed the Service Request & Consultation Lifecycle state model, and authored the 20 Major Use Cases.
Daksh Mittal (24UCS002)	Authored the 3 Detailed Use Cases, formulated Agile User Stories (US-001 to US-015), defined Acceptance Criteria, MoSCoW Prioritization, and the Traceability Matrix (RTM).

30. AI Usage Policy – Student Declaration

AI tools may be used as supporting tools for this assignment, but the submitted work must represent the team's own understanding and contribution.
We declare that:

- We have disclosed the use of AI tools, if any, during preparation of this assignment.
- AI-generated suggestions have been reviewed, verified, edited and adapted by the team.
- We understand and can explain all submitted requirements, use cases, business rules, and user stories during a viva/discussion.
- We have verified that the requirements are consistent with our project's scope and objectives.
- The final SRS represents our team's own understanding and decisions.

Field	Declaration Details
AI Tool(s) Used:	Antigravity IDE (Gemini 3.7 Flash)
Purpose of AI Use:	Requirement synthesis, structuring SRS v1.0 specifications, format alignment with course template, and diagram drafting
Team ID:	T039
Team Member Names:	Mehul Baid (24UCS003), Alakh Agarwal (24UCS001), Yug Nahar (24UCS004), Daksh Mittal (24UCS002)
Date:	September 7, 2026

Team Declaration:
We confirm that the information provided above is accurate and that we accept responsibility for the submitted work.
Team Representative Signature/Confirmation (Write full names in Uppercase only):
MEHUL BAID, ALAKH AGARWAL, YUG NAHAR, DAKSH MITTAL

Submission Note: This document serves as the formal SRS v1.0 baseline for subsequent lab activities and will be used while preparing UML analysis diagrams, architectural system design, full-stack implementation, and verification test suites.

The LNM Institute of Information Technology, Jaipur
Software Development Laboratory
Course Code: CS314
Lab Assignment 3: Solution
Scenario, Use Case and Activity Modelling

Item	Details
Project Name	Student Wellness Portal
Team ID	T039
Team Members	Mehul Baid, Alakh Agarwal, Yug Nahar, Daksh Mittal
Roll Numbers	24UCS003, 24UCS001, 24UCS004, 24UCS002
Detailed Use Cases Carried Forward	UC-005, UC-007, UC-010 (same as Assignment 2)

Objective

This document extends the Requirements Specification prepared in Assignment 2 by developing scenarios and UML models — a Use Case Diagram and three Activity Diagrams — for the Student Wellness Portal. As required, the same three detailed use cases carried forward from Assignment 2 are used: UC-005 (Submit Confidential Wellness Support Request), UC-007 (Book Counseling Consultation Appointment), and UC-010 (Conduct Clinical Triage & Case Assignment). No new functionality has been introduced beyond what is already specified in the Assignment 2 requirements and use cases.

1. Scenarios

For each of the three detailed use cases carried forward from Assignment 2, a brief scenario is presented below describing the actor(s) involved, the starting condition, the main sequence of activities, an important alternative/exception flow, and the expected outcome.

Scenario 1 — UC-005: Submit Confidential Wellness Support Request

Actor(s) Involved: Student (primary actor); System (Student Wellness Portal).
Starting Condition: An authenticated student has fewer than two active open support requests and wishes to seek confidential help for a personal concern.
Main Sequence of Activities:

1. Student navigates to the Wellness Services Catalog and selects a relevant service, then chooses 'Request Support'.
2. System displays a multi-step confidential request form.
3. Student selects an urgency rating (Routine, Priority, or Urgent) and a preferred consultation mode (In-person or Telehealth).
4. Student enters a background description and preferred availability.
5. Student reviews the confidentiality notice and submits the request.
6. System validates the active-request limit and the submitted form inputs.
7. System persists the request with status PENDING REVIEW and alerts the counselor triage queue.

Alternative / Exception Flow: If the student already has two pending requests, the system blocks the submission and notifies the student to resolve the existing request(s) first — no

request record is created in this case. If the urgency rating is 'Urgent', the system additionally displays 24/7 crisis helpline numbers and routes the ticket directly to high-priority triage.

Expected Outcome: A valid confidential service-request record is created with status PENDING REVIEW (or the routing described above for urgent cases), and the student receives an in-app confirmation with a unique Request ID.

Scenario 2 — UC-007: Book Counseling Consultation Appointment

Actor(s) Involved: Student (primary actor); System (Student Wellness Portal).
Starting Condition: The student's support request has already been triaged and the assigned counselor has open availability slots.
Main Sequence of Activities:

1. Student opens the triaged support request and clicks 'Schedule Appointment'.
2. System retrieves and renders the assigned counselor's calendar of available slots.
3. Student selects a desired date and a 45-minute time slot.
4. System verifies slot availability and enforces the 24-hour advance-booking rule for routine requests.
5. Student confirms the appointment mode (In-person or Telehealth) and submits the booking.
6. System atomically locks the slot and creates an Appointment record with status CONFIRMED.
7. If Telehealth mode was selected, the system generates a secure virtual consultation URL.
8. System updates the associated ServiceRequest status to SCHEDULED and sends confirmation notifications to the student and counselor.

Alternative / Exception Flow: If the selected slot is taken concurrently by another booking, the system aborts the lock attempt and refreshes the list of open slots for the student to choose again. If the student attempts to book a routine appointment with less than 24 hours' notice, the system restricts the booking and notifies the student of the advance-notice requirement.

Expected Outcome: A confirmed Appointment record is created, the corresponding calendar slot is locked against double-booking, a Telehealth URL is attached where applicable, and both participants are notified.

Scenario 3 — UC-010: Conduct Clinical Triage & Case Assignment

Actor(s) Involved: Wellness Staff / Counselor (primary actor); System (Student Wellness Portal); Campus Health Center (secondary actor, for the escalation path).
Starting Condition: A counselor with elevated privileges is authenticated, and one or more student support requests exist in PENDING REVIEW status.
Main Sequence of Activities:

1. Counselor accesses the Staff Triage Dashboard.
2. System presents incoming requests sorted by urgency and submission timestamp.
3. Counselor reviews the student's background concerns, urgency rating, and request history.
4. Counselor assigns a clinical priority level and allocates the case to a counselor.
5. Counselor submits the triage evaluation.
6. System updates the ServiceRequest status to TRIAGED, notifies the student, and unlocks the appointment-booking calendar.
7. System logs the triage action in the tamper-evident audit trail.

Alternative / Exception Flow: If the counselor identifies an acute self-harm risk or psychiatric emergency during review, the counselor triggers the emergency escalation path (UC-011) instead

of routine triage: the system updates the request status to ESCALATED and alerts the Campus Health Center, which acknowledges the referral and coordinates immediate intervention. This escalation path bypasses standard slot booking, and the escalation is separately logged in the audit trail.

Expected Outcome: The ServiceRequest status is updated to TRIAGED (routine/priority path) or ESCALATED (emergency path), the student is notified accordingly, and an audit-trail entry is recorded.

2. Use Case Diagram

The diagram below presents the complete UML Use Case Diagram for the Student Wellness Portal. It shows all actors (Student, Wellness Staff / Counselor, Administrator, Institute Management, and Health Center / Medical Referral Specialist), the full set of use cases from the Assignment 2 requirements, actor–use case associations, «include» and «extend» relationships where justified, and a generalization relationship (Manage Appointment generalizing Reschedule Appointment and Cancel Appointment). The three detailed use cases carried forward from Assignment 2 — UC-005, UC-007, and UC-010 — are highlighted with a bold outline and a star marker, as shown in the legend.

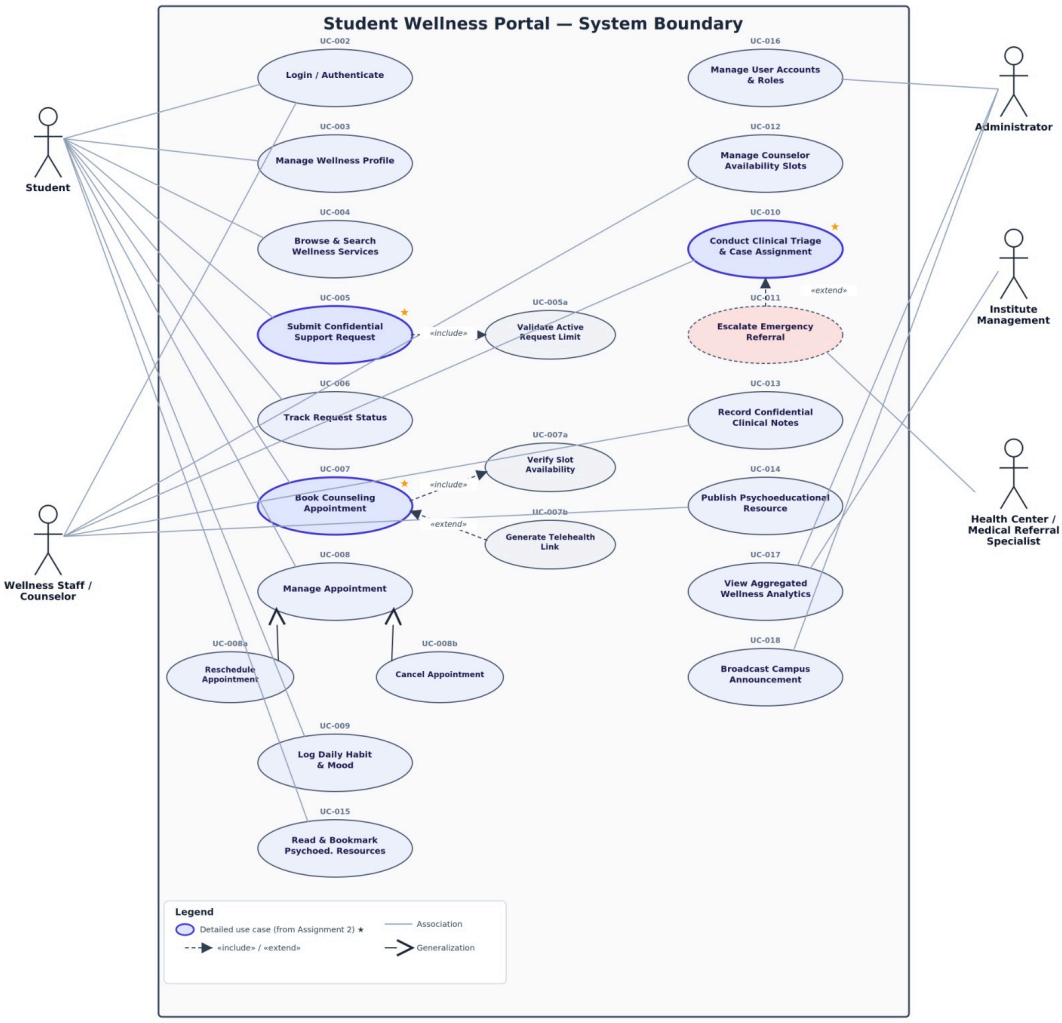


Figure 2.1: UML Use Case Diagram — Student Wellness Portal

3. Activity Diagrams

An Activity Diagram is provided for each of the same three detailed use cases, showing initial and final nodes, activities, control flow, decision/merge points, and swimlanes for the actors and system components involved.

3.1 Activity Diagram — UC-005: Submit Confidential Wellness Support Request

Swimlanes: Student and **System (Student Wellness Portal)**. The flow captures the multi-step confidential request form, the 'active request limit' validation with a blocking exception path, and the branch on urgency rating that routes 'Urgent' submissions to the 24/7 crisis helpline display and high-priority triage, while Routine/Priority submissions generate a standard confirmation and alert the counselor triage queue.

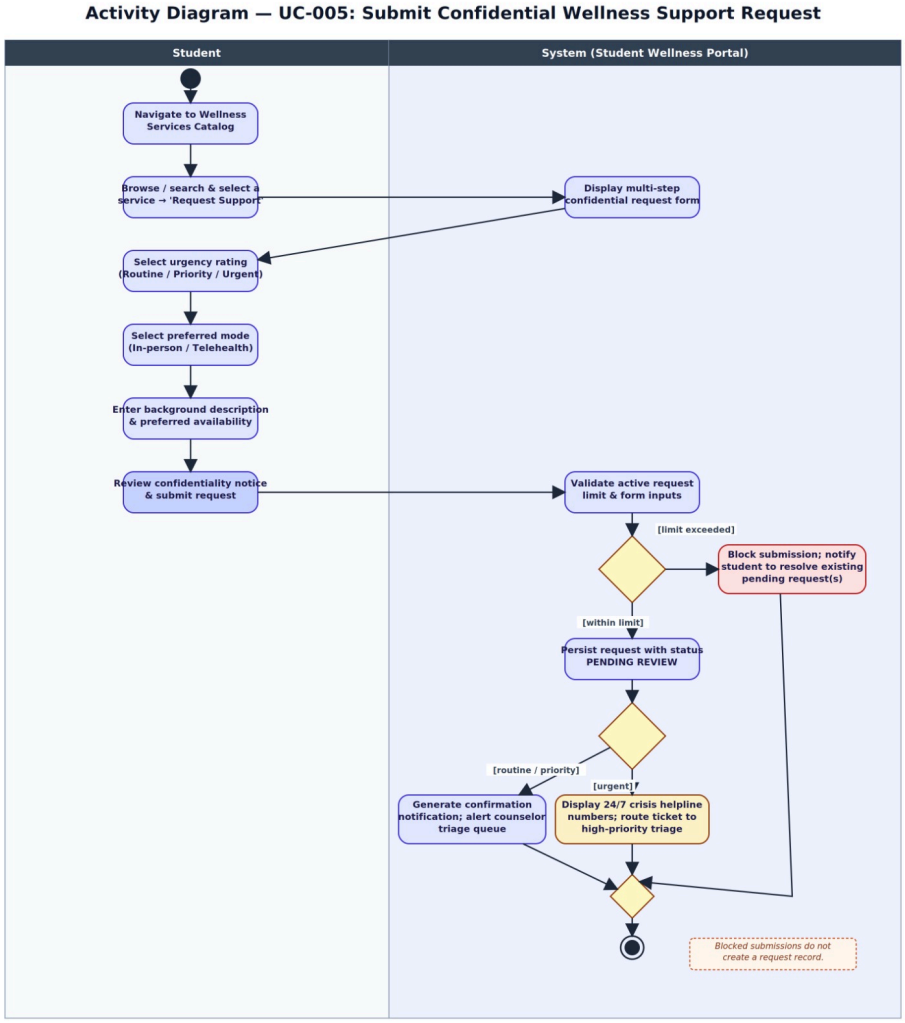


Figure 3.1: Activity Diagram — UC-005

3.2 Activity Diagram — UC-007: Book Counseling Consultation Appointment

Swimlanes: Student and **System (Student Wellness Portal)**. The diagram models slot retrieval and selection, the availability/24-hour-advance-rule check with a concurrency-abort exception

path and a ‘less-than-24-hour notice’ restriction path, the atomic slot lock and appointment creation, and a decision node for Telehealth-URL generation («extend» UC-007b) before the ServiceRequest status is updated and confirmations are sent.

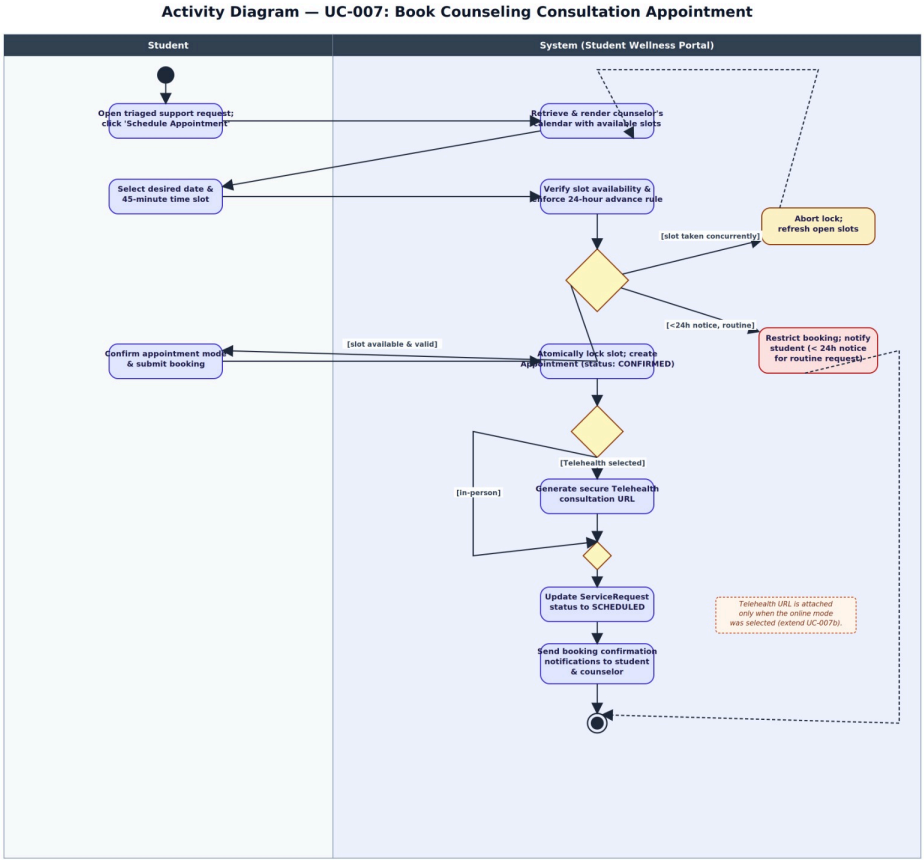


Figure 3.2: Activity Diagram — UC-007

3.3 Activity Diagram — UC-010: Conduct Clinical Triage & Case Assignment

Swimlanes: Wellness Staff / Counselor, **System (Student Wellness Portal)**, and **Campus Health Center**. The diagram shows the counselor’s review and triage-evaluation activities, and a decision node branching on case severity: the routine/priority path updates the request to TRIAGED and unlocks appointment booking, while the acute self-harm/emergency path escalates the case to the Campus Health Center swimlane, which acknowledges the referral and coordinates intervention. Both paths conclude with an entry in the tamper-evident audit trail.

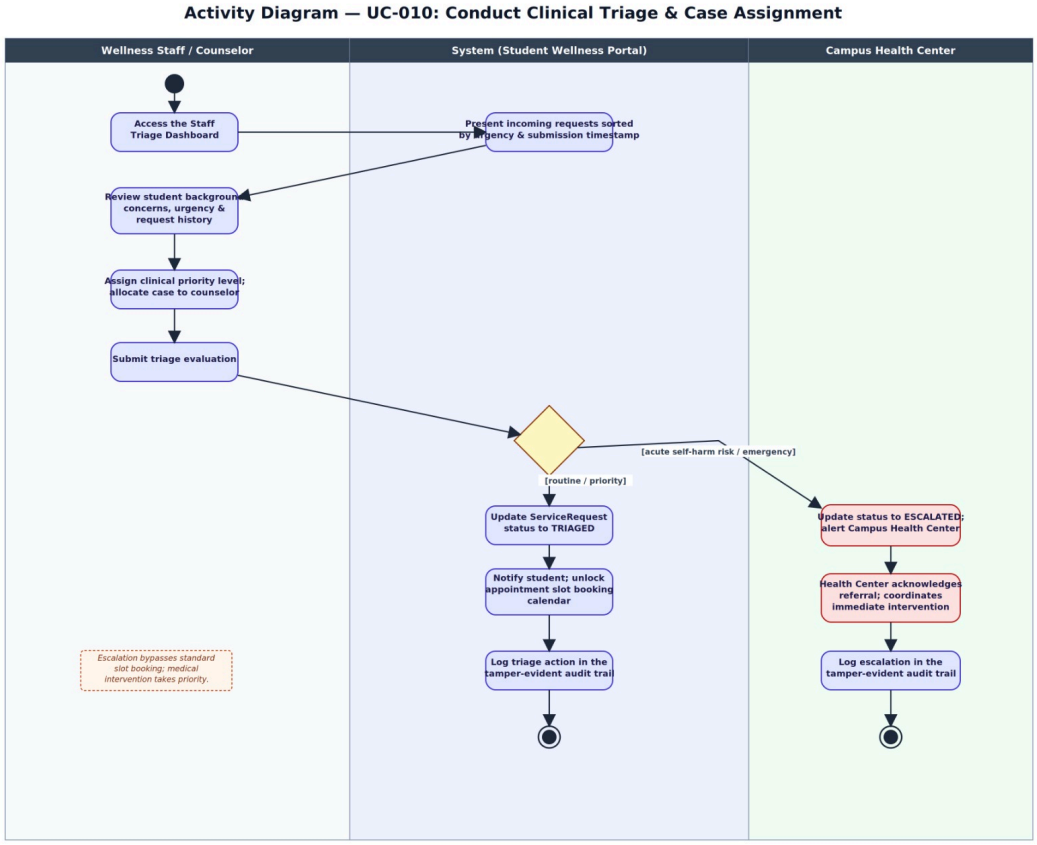


Figure 3.3: Activity Diagram — UC-010

Consistency Requirement

The following relationship has been maintained throughout this assignment, with no new functionality introduced beyond the Assignment 2 baseline:

Assignment 2 Requirements → 3 Detailed Use Cases → Scenarios → Use Case Diagram → Activity Diagrams

Each scenario in Section 1 traces directly to its corresponding detailed use case in the Assignment 2 SRS (UC-005, UC-007, UC-010); the Use Case Diagram in Section 2 reflects the same 20 use cases, actors, and business rules defined in Assignment 2 (e.g., BR-002 active-request limit, BR-003 24-hour advance rule, BR-007 emergency escalation); and the Activity Diagrams in Section 3 model the exact main and alternative flows described in the Assignment 2 detailed use cases and the scenarios above.

Contributions

The individual contribution of each team member towards this assignment is recorded below:

Student Name & Roll No.	Individual Contribution (Assignment 3)
Yug Nahar (24UCS004)	Authored the Scenario and Activity Diagram for UC-007 (Book Counseling Consultation Appointment), and prepared the complete UML Use Case Diagram for the system.

Mehul Baid (24UCS003)	Authored the Scenario and Activity Diagram for UC-005 (Submit Confidential Wellness Support Request).
Alakh Agarwal (24UCS001)	Authored the Scenario and Activity Diagram for UC-010 (Conduct Clinical Triage & Case Assignment).
Daksh Mittal (24UCS002)	Reviewed all three scenarios and activity diagrams for consistency with the Assignment 2 requirements, and compiled the final document.

AI Usage Policy – Student Declaration

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We declare that:

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- AI-generated suggestions have been reviewed, verified, edited and adapted by the team.
- We understand and can explain all submitted requirements, use cases, business rules, and user stories during a viva/discussion.
- We have verified that the requirements are consistent with our project's scope and objectives.
- The final SRS represents our team's own understanding and decisions.

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Team ID:	T039
Team Member Names:	Mehul Baid (24UCS003), Alakh Agarwal (24UCS001), Yug Nahar (24UCS004), Daksh Mittal (24UCS002)
Date:	September 7, 2026

Team Declaration:

We confirm that the information provided above is accurate and that we accept responsibility for the submitted work.

Team Representative Signature/Confirmation (Write full names in Uppercase only):

MEHUL BAID, ALAKH AGARWAL, YUG NAHAR, DAKSH MITTAL

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Software Development Laboratory
Course Code: CS314

Lab Assignment 4
Classes, Object Diagram and Relationships

1. Classes — Class Identification

The following classes are derived from the same three detailed use cases carried forward from Assignment 2 and the corresponding Activity Diagrams in Assignment 3. The model uses 12 meaningful classes; Student and Counselor specialize the common User responsibilities.

Class	Important attributes	Operations / responsibilities	Traceability
User	userId; fullName; email; role; accountStatus	authenticate(); logout(); updateProfile()	UC-005, UC-007, UC-010 preconditions
Student	rollNumber; department; contactNumber	submitSupportRequest(); bookAppointment(); receiveNotification()	UC-005, UC-007
Counselor	counselorId; specialization	triageRequest(); assignCase(); manageSlots()	UC-007, UC-010
WellnessProfile	profileId; primaryConcerns; preferredMode; confidentialityPreference	updateProfile(); updatePreferences()	UC-005/UC-010 student context
WellnessService	serviceId; title; category; mode; status	viewDetails(); checkStatus()	UC-005 selected active service

Class-modeling decisions

• **User** is modeled as an abstract superclass because authentication, account status and common profile operations apply across roles.
• **Student** and **Counselor** are role-specific specializations rather than treating every actor as an independent unrelated class.
• **WellnessService**, **ServiceRequest** and **WellnessProfile** directly represent domain information already defined in the Assignment 2 data requirements.

The LNM Institute of Information Technology, Jaipur
Software Development Laboratory
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Lab Assignment 4
Classes, Object Diagram and Relationships

1. Classes — Class Identification (continued)

Class	Important attributes	Operations / responsibilities	Traceability
ServiceRequest	requestId; urgencyLevel; description; preferredSchedule; status	submitRequest(); validate(); updateStatus(); assignCounselor()	UC-005; UC-007; UC-010
TriageEvaluation	evaluationId; priorityLevel; remarks; evaluatedAt	evaluate(); assignPriority()	UC-010
AvailabilitySlot	slotId; date; startTime; endTime; status	checkAvailability(); lockSlot(); releaseSlot()	UC-007
Appointment	appointmentId; date; startTime; endTime; mode; status	book(); confirm(); cancel(); reschedule()	UC-007
Notification	notificationId; type; title; message; isRead	send(); markAsRead()	UC-005; UC-007; UC-010
AdminAuditLog	auditId; actionType; targetEntity; timestamp; details	recordAction()	UC-010 audit trail
HealthCenterReferral	referralId; reason; priority; referralStatus	createReferral(); updateStatus()	UC-010 emergency alternative flow

Use-case traceability

Detailed use case	Classes directly supporting it
UC-005 — Submit Confidential Wellness Support Request	Student, User, WellnessProfile, WellnessService, ServiceRequest, Notification
UC-007 — Book Counseling Consultation Appointment	Student, User, ServiceRequest, Counselor, AvailabilitySlot, Appointment, Notification
UC-010 — Conduct Clinical Triage & Case Assignment	Counselor, User, ServiceRequest, TriageEvaluation, Notification, AdminAuditLog, HealthCenterReferral

The class set is deliberately focused on the three selected use cases and their documented alternative/exception flows. No unrelated project modules such as habit analytics or psychoeducational resources are introduced into this Assignment 4 model.

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Lab Assignment 4
Classes, Object Diagram and Relationships

3. UML Class Diagram

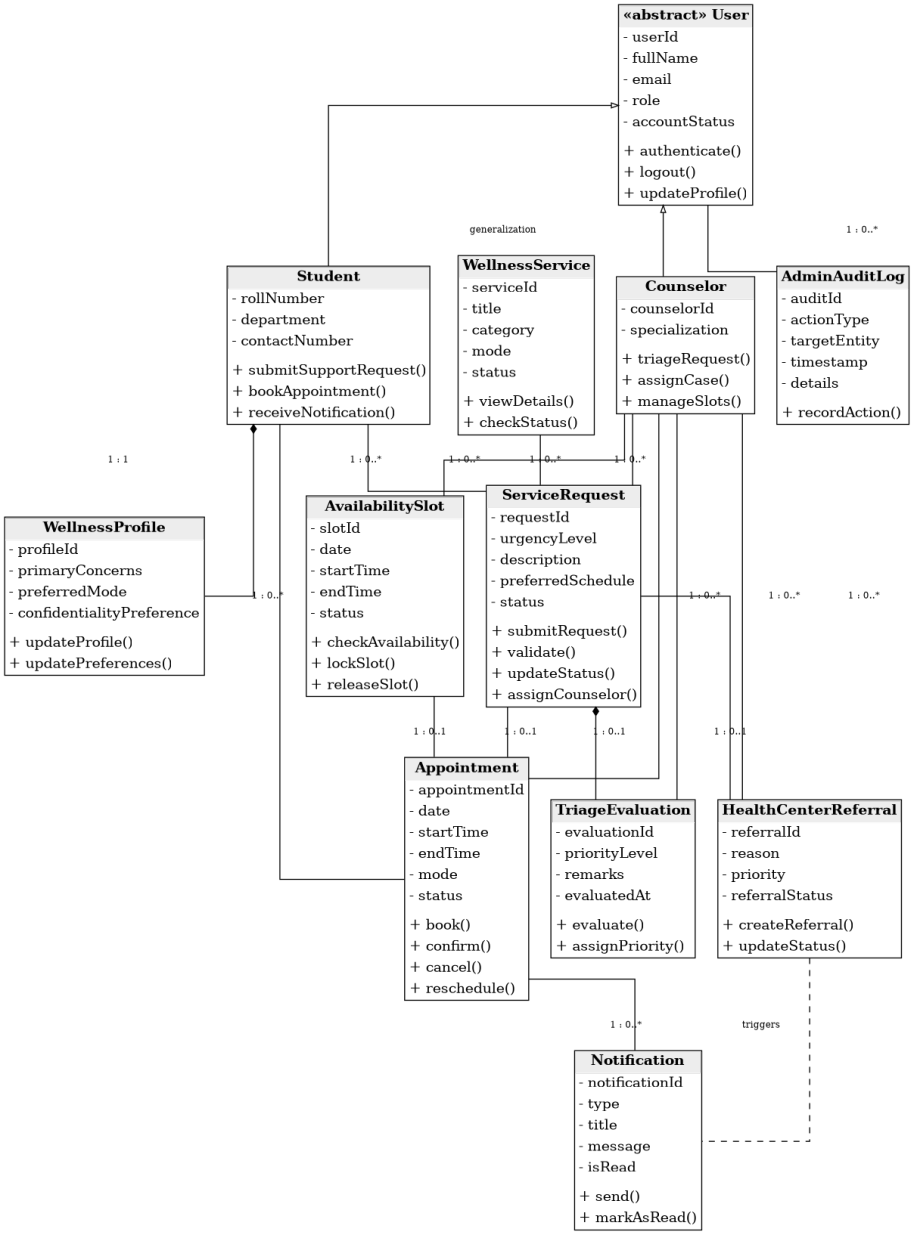
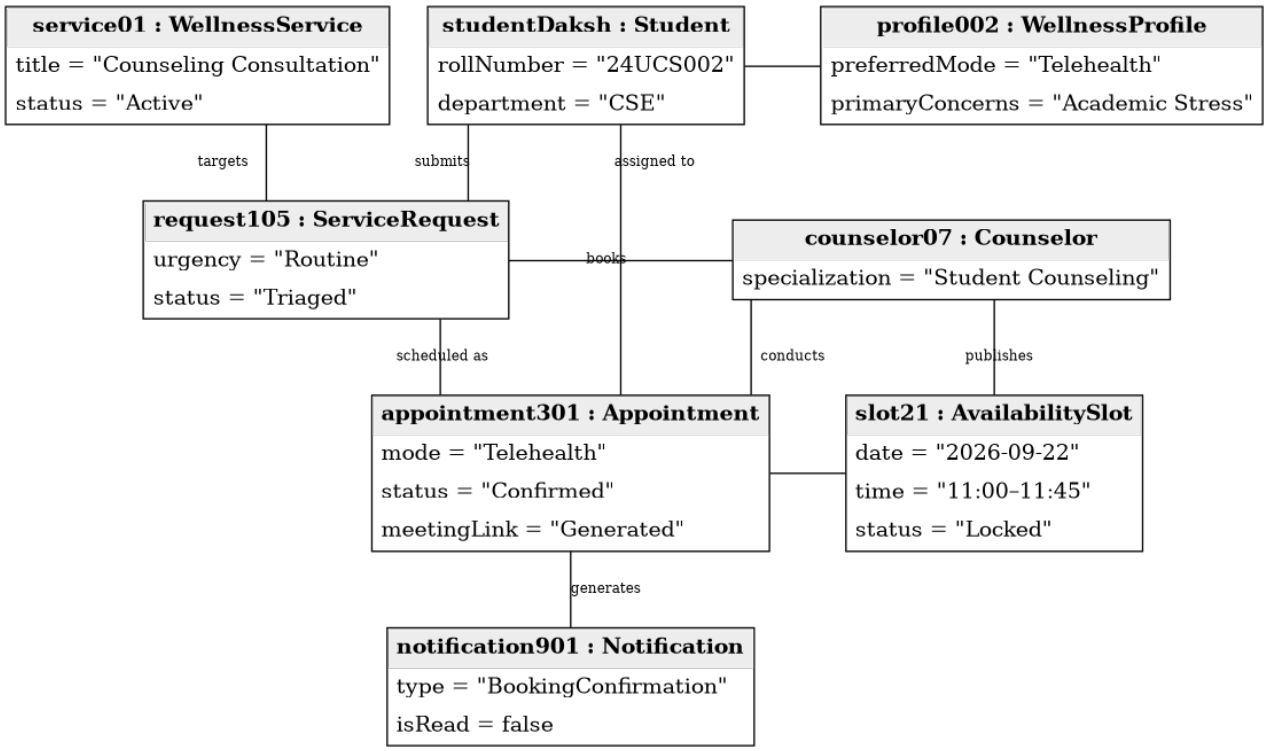


Figure 4.1: Class Diagram for UC-005, UC-007 and UC-010

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Lab Assignment 4
Classes, Object Diagram and Relationships

2. UML Object Diagram



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Lab Assignment 4
Classes, Object Diagram and Relationships

2. Object Diagram — Scenario and Object Details

Selected use case: UC-007 — Book Counseling Consultation Appointment

Scenario description (4 sentences): The object snapshot represents a student whose support request has already been triaged and who is now booking a counseling consultation. The selected wellness service is active, the counselor has published an available 45-minute slot, and the student confirms a Telehealth appointment. The system locks the selected availability slot and creates a confirmed Appointment associated with the request and counselor. A booking-confirmation Notification is generated after successful booking.

Representative objects and values

Object / instance	Representative attribute values
studentDaksh : Student	rollNumber = "24UCS002"; department = "CSE"
profile002 : WellnessProfile	preferredMode = "Telehealth"; primaryConcerns = "Academic Stress"
service01 : WellnessService	title = "Counseling Consultation"; status = "Active"
request105 : ServiceRequest	urgency = "Routine"; status = "Triaged"
counselor07 : Counselor	specialization = "Student Counseling"
slot21 : AvailabilitySlot	date = "2026-09-22"; time = "11:00–11:45"; status = "Locked"
appointment301 : Appointment	mode = "Telehealth"; status = "Confirmed"; meetingLink = "Generated"
notification901 : Notification	type = "BookingConfirmation"; isRead = false

Object-model consistency: every object shown here is an instance of a class identified in the class model on pages 33–34. The snapshot represents one meaningful point in the UC-007 lifecycle rather than the complete system state.

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Lab Assignment 4
Classes, Object Diagram and Relationships

4. Relationships — Documentation

The following relationships are justified using the requirements, business rules and the three detailed use cases. Multiplicities represent the allowed associations in the model rather than the temporary values of one object snapshot.

Relationship	Type	Multiplicity	Justification
User → Student / Counselor	Generalization / Inheritance	1 → many subclasses	Student and Counselor share authentication/account responsibilities while adding role-specific responsibilities.
Student → WellnessProfile	Composition	1 : 1	A student's wellness profile belongs to that student and supports confidential profile/context data.
Student → ServiceRequest	Association	1 : 0..*	A student can submit multiple requests over time; BR-002 limits active open requests to two.
WellnessService → ServiceRequest	Association	1 : 0..*	A request is submitted for one active wellness service, while a service can receive many requests.
Counselor → ServiceRequest	Association	1 : 0..*	A counselor may be assigned multiple support cases during triage and case management.
ServiceRequest → TriageEvaluation	Composition	1 : 0..1	A triage evaluation is created for a request when the counselor evaluates and assigns priority.
Counselor → TriageEvaluation	Association	1 : 0..*	Counselors perform triage evaluations for incoming requests.

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Lab Assignment 4
Classes, Object Diagram and Relationships

4. Relationships — Continued & Consistency

Relationship	Type	Multiplicity	Justification
Counselor → AvailabilitySlot	Association	1 : 0..*	Counselors publish and manage weekly consultation slots.
ServiceRequest → Appointment	Association	1 : 0..1	A triaged request may produce an appointment; UC-007 changes the request to SCHEDULED.
AvailabilitySlot → Appointment	Association	1 : 0..1	One slot can be locked by at most one confirmed appointment, preventing double-booking.
Student → Appointment	Association	1 : 0..*	Students can book counseling appointments associated with their triaged requests.
Counselor → Appointment	Association	1 : 0..*	A counselor can conduct multiple appointments across available slots.
Appointment → Notification	Association	1 : 0..*	Confirmed bookings trigger confirmation/reminder notifications.
User → AdminAuditLog	Association	1 : 0..*	Authenticated actions such as triage assignment are recorded in the audit trail.
ServiceRequest → HealthCenterReferral	Association	1 : 0..1	An urgent/emergency case can be escalated to the campus health center.
Counselor → HealthCenterReferral	Association	1 : 0..*	The counselor initiates emergency referrals when the alternative flow requires escalation.
HealthCenterReferral → Notification	Dependency / Association	1 : 0..*	Emergency referral creation triggers alerts to relevant users/health-center staff.

Consistency chain

Assignment 2 Requirements → 3 Detailed Use Cases → Assignment 3 Activities → Classes → Objects → Relationships

Assignment 2 requirement / rule	Use case	Assignment 4 model evidence
FR-024/025: structured request and urgency	UC-005	ServiceRequest.urgencyLevel, description, submitRequest()
FR-033/034/035: slot-based booking and double-booking prevention	UC-007	AvailabilitySlot, Appointment, lockSlot(), 1:0..1 slot→appointment
FR-029/042: request assignment and triage queue	UC-010	Counselor, TriageEvaluation, ServiceRequest.assigned counselor
FR-064: event notifications	UC-005/007/010	Notification associated with request/appointment/escalation events
FR-032 / NFR-007: timestamped audit trail	UC-010	AdminAuditLog linked to User and triage actions

The model therefore preserves the required traceability instead of introducing unrelated classes or relationships.

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Lab Assignment 4
Classes, Object Diagram and Relationships

5. Contributions & AI Usage Declaration

Contributions

Team member	Assignment 4 contribution
Mehul Baid (24UCS003)	Class identification and class descriptions; reviewed ServiceRequest and WellnessService modeling.
Alakh Agarwal (24UCS001)	Relationship identification and triage/emergency modeling; reviewed multiplicities.
Yug Nahar (24UCS004)	Object-model scenario and Object Diagram preparation for UC-007.
Daksh Mittal (24UCS002)	Consistency verification, documentation compilation, formatting and final submission preparation.

AI USAGE POLICY – STUDENT DECLARATION

AI Tool(s) Used: **ChatGPT**
Purpose of AI Use: **Supporting class identification, UML relationship drafting, object-snapshot modeling, and document formatting based on the team's existing requirements and use cases.**
Team ID: **T039**
Team Representative: **MEHUL BAID**
Date: **September 20, 2026**

Team declaration: The generated suggestions must be reviewed, verified and understood by the team before submission. The final UML model is intended to remain consistent with the Assignment 2 requirements, the three selected use cases, and the Assignment 3 activity diagrams.

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Lab Assignment 5
Collaboration, State and Component Modelling

1. Project Identification

Item	Details
Project Name	Student Wellness Portal
Project Type	Web-based Institutional Wellness, Counseling & Activity Tracking Platform
Team ID	T039
Team Members	Mehul Baid (24UCS003), Alakh Agarwal (24UCS001), Yug Nahar (24UCS004), Daksh Mittal (24UCS002)
Development Model	Agile + Incremental Development Framework
Primary Users	Students, Wellness Staff / Counselors, Administrators, Health Center Staff
Course	Software Engineering / Software Development Laboratory (CS314)
Detailed Use Cases Carried Forward	UC-005 (Submit Confidential Support Request), UC-007 (Book Appointment), UC-010 (Conduct Clinical Triage)

2. Executive Objective & Modelling Scope

This document completes Lab Assignment 5 by extending the structural models formulated in Assignment 4 (Classes, Object Diagram, and Class Relationships) into behavioral interaction models, object state machine models, and software component architecture diagrams for the Student Wellness Portal. In accordance with course directives, strict end-to-end consistency is maintained across the complete software engineering chain: *Requirements (Assignment 2) → 3 Detailed Use Cases → Scenarios & Activity Diagrams (Assignment 3) → Classes & Objects (Assignment 4) → Collaboration Diagrams → State Diagrams → Component Architecture (Assignment 5).*

3. Summary of Carried-Forward Domain Classes

The interaction and state models in this assignment directly consume the 12 domain classes established in Assignment 4:

Class Name	Key Attributes	Primary Responsibilities
`User (Abstract)`	userId, fullName, email, role, accountStatus	Authentication, session control, role-based access control (RBAC).
`Student`	rollNumber, department, contactNumber	Submit confidential support requests, book consultation slots, view habit logs.
`Counselor`	counselorId, specialization	Triage support cases, assign clinical priority, manage consultation availability slots.
`WellnessProfile`	profileId, primaryConcerns, preferredMode, confidentialityPref	Maintain student emergency contacts, preferences, and anonymity settings.
`WellnessService`	serviceId, title, category, mode, status	Maintain active institutional wellness service catalog entries.
`ServiceRequest`	requestId, urgencyLevel, description, status	Manage student support request lifecycle across triage, booking, and closure.
`AvailabilitySlot`	slotId, date, startTime, endTime, status	Slot availability management, capacity tracking, and atomic locking.
`Appointment`	appointmentId, mode, meetingLink, status	Confirmed counseling consultation bookings, Telehealth link binding.
`TriageEvaluation`	evaluationId, priorityLevel, remarks	Clinical assessment notes, risk priority records, and triage remarks.
`HealthCenterReferral`	referralId, reason, priority, referralStatus	Emergency medical escalation protocol dispatch (BR-007 compliance).
`Notification`	notificationId, type, message, isRead	In-app real-time alerts and SMTP email notification dispatch.
`AdminAuditLog`	auditId, actionType, timestamp, details	Immutable system-wide audit logging for clinical and admin actions.

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Lab Assignment 5
Collaboration, State and Component Modelling

4. Task 1: UML Collaboration Diagrams (Communication Diagrams)

Collaboration Diagrams (UML 1.x / Communication Diagrams) model the structural organization of participating objects, their static links, and the sequenced messages exchanged to fulfill each detailed use case.

4.1 Collaboration Diagram — UC-005: Submit Confidential Wellness Support Request

Scenario Summary: An authenticated student selects an active wellness service, retrieves confidentiality preferences from their wellness profile, and submits a structured support request. The system validates the active request limit (BR-002: maximum 2 active requests), sets the request status to PENDING REVIEW, and dispatches an in-app confirmation notification.

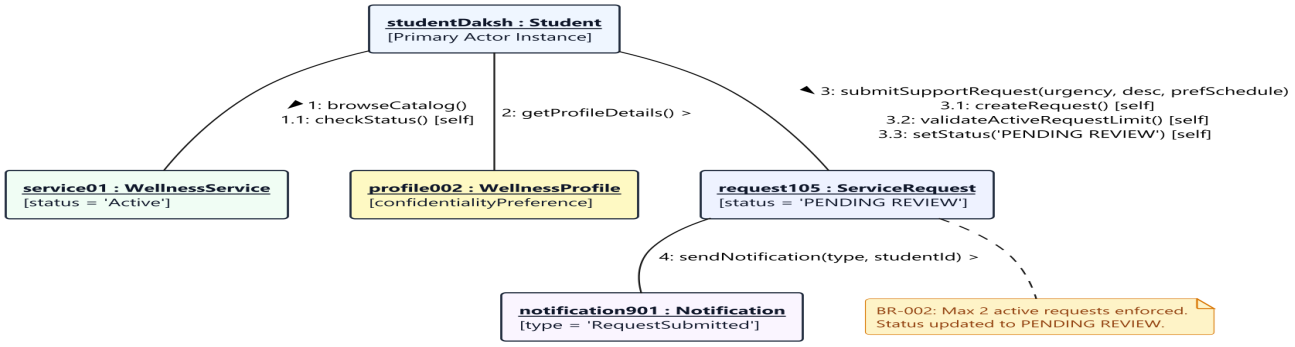


Figure 5.1: UML Collaboration Diagram for UC-005 (Submit Support Request)

Interaction & Responsibility Specification Table (UC-005)

Seq #	Sender Object	Receiver Object	Message Signature & Parameters	Object Responsibility / Semantic Output
1	`studentDaksh`	`service01`	`browseCatalog()`	Browses active campus wellness service categories.
1.1	`service01`	`service01`	`checkStatus()`	Verifies operational status is Active (self-call).
2	`studentDaksh`	`profile002`	`getProfileDetails()`	Fetches emergency contacts and confidentiality settings.
3	`studentDaksh`	`request105`	`submitSupportRequest(urgency, desc, prefSchedule)`	Initiates confidential support request submission form.
3.1	`request105`	`request105`	`createRequest()`	Instantiates new request entity with user payload (self).
3.2	`request105`	`request105`	`validateActiveRequestLimit()`	Enforces BR-002: blocks submission if active open requests ≥ 2.
3.3	`request105`	`request105`	`setStatus('PENDING REVIEW')`	Updates status to PENDING REVIEW and enqueues ticket (self).
4	`request105`	`notification901`	`sendNotification(type, studentId)`	Dispatches in-app confirmation notification with unique RequestID.

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Lab Assignment 5
Collaboration, State and Component Modelling

4.2 Collaboration Diagram — UC-007: Book Counseling Consultation Appointment

Scenario Summary: An eligible student with a triaged support request browses their assigned counselor's open slots, selects a 45-minute consultation slot, and confirms the booking mode (Telehealth). The system validates the 24-hour advance booking rule (BR-003), locks the availability slot atomically (BR-004), generates a secure Telehealth URL (BR-011), updates the request status to SCHEDULED, and dispatches automated booking notifications.

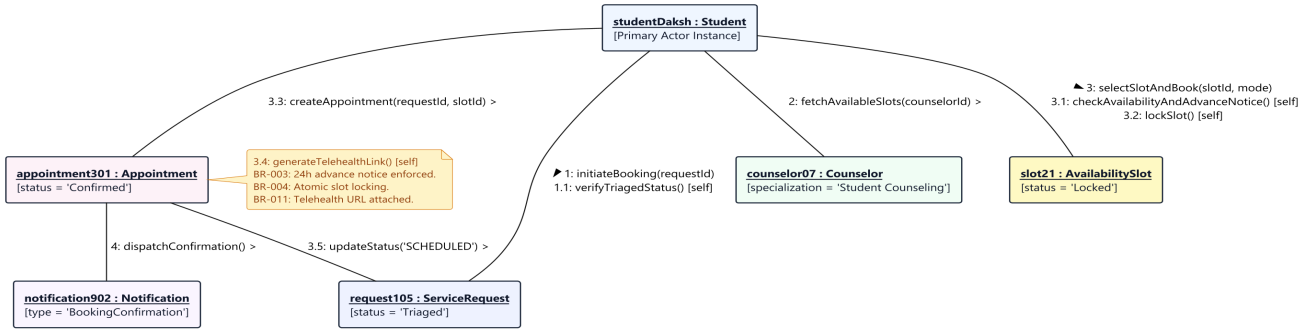


Figure 5.2: UML Collaboration Diagram for UC-007 (Book Appointment)

Interaction & Responsibility Specification Table (UC-007)

Seq #	Sender Object	Receiver Object	Message Signature & Parameters	Object Responsibility / Semantic Output
1	`studentDaksh`	`request105`	`initiateBooking(requestId)`	Selects approved/triaged support request for appointment scheduling.
1.1	`request105`	`request105`	`verifyTriagedStatus()`	Confirms request status is TRIAGED prior to opening calendar (self).
2	`studentDaksh`	`counselor07`	`fetchAvailableSlots(counselorId)`	Queries assigned counselor's published open consultation calendar.
3	`studentDaksh`	`slot21`	`selectSlotAndBook(slotId, mode)`	Selects desired date and 45-minute consultation slot.
3.1	`slot21`	`slot21`	`checkAvailabilityAndAdvanceNotice()`	Enforces BR-003: validates minimum 24-hour advance booking notice.
3.2	`slot21`	`slot21`	`lockSlot()`	Atomically locks slot against double-booking race conditions (BR-004).
3.3	`studentDaksh`	`appointment301`	`createAppointment(requestId, slotId)`	Instantiates confirmed Appointment record with status CONFIRMED.
3.4	`appointment301`	`appointment301`	`generateTelehealthLink()`	Dynamically attaches secure virtual meeting URL (BR-011) (self).
3.5	`appointment301`	`request105`	`updateStatus('SCHEDULED')`	Transitions associated ServiceRequest status from TRIAGED to SCHEDULED.
4	`appointment301`	`notification902`	`dispatchConfirmation()`	Dispatches automated booking confirmation alerts to student and counselor.

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Lab Assignment 5
Collaboration, State and Component Modelling

4.3 Collaboration Diagram — UC-010: Conduct Clinical Triage & Case Assignment

Scenario Summary: An authorized counselor accesses the staff triage dashboard, reviews an incoming support request, and records clinical triage notes. For routine cases, the counselor assigns clinical priority and unlocks appointment booking. If an acute psychiatric risk is identified, the counselor triggers emergency escalation (BR-007), generating a referral to the Campus Health Center and recording a tamper-evident audit log.

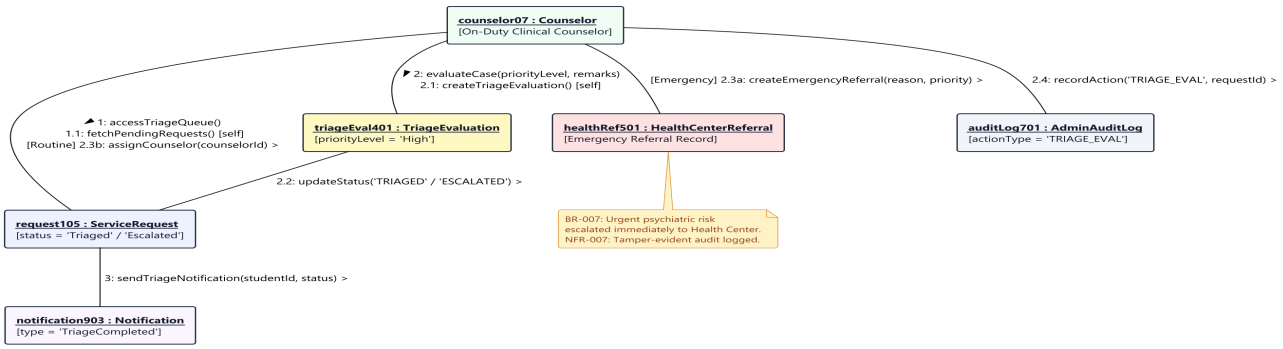


Figure 5.3: UML Collaboration Diagram for UC-010 (Clinical Triage & Case Assignment)

Interaction & Responsibility Specification Table (UC-010)

Seq #	Sender Object	Receiver Object	Message Signature & Parameters	Object Responsibility / Semantic Output
1	`counselor07`	`request105`	`accessTriageQueue()`	Opens counselor clinical triage dashboard and active queue.
1.1	`request105`	`request105`	`fetchPendingRequests()`	Retrieves pending requests sorted by urgency and timestamp (self).
2	`counselor07`	`triageEval401`	`evaluateCase(priorityLevel, remarks)`	Inputs clinical assessment notes and determines care path.
2.1	`triageEval401`	`triageEval401`	`createTriageEvaluation()`	Persists evaluation record linked to support request (self).
2.2	`triageEval401`	`request105`	`updateStatus('TRIAGED' / 'ESCALATED')`	Updates request lifecycle state according to clinical severity.
2.3a	`counselor07`	`healthRef501`	`createEmergencyReferral(reason, priority)`	**[Emergency Flow]:** Dispatches urgent Campus Health Center referral.
2.3b	`counselor07`	`request105`	`assignCounselor(counselorId)`	**[Routine Flow]:** Binds assigned qualified counselor to support case.
2.4	`counselor07`	`auditLog701`	`recordAction('TRIAGE_EVAL', requestId)`	Logs immutable, tamper-evident audit record of triage decision (NFR-007).
3	`request105`	`notification903`	`sendTriageNotification(studentId, status)`	Notifies student and unlocks slot booking calendar upon routine triage.

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Lab Assignment 5
Collaboration, State and Component Modelling

5. Task 2: UML State Machine Diagrams

State Machine Diagrams model the internal lifecycle states, triggering events, transition paths, and guard conditions for key domain entities in the Student Wellness Portal.

5.1 State Machine Diagram — Entity 1: ServiceRequest

The **ServiceRequest** entity models the complete lifecycle of a student's support submission across triage evaluation, appointment scheduling, consultation execution, and clinical resolution.

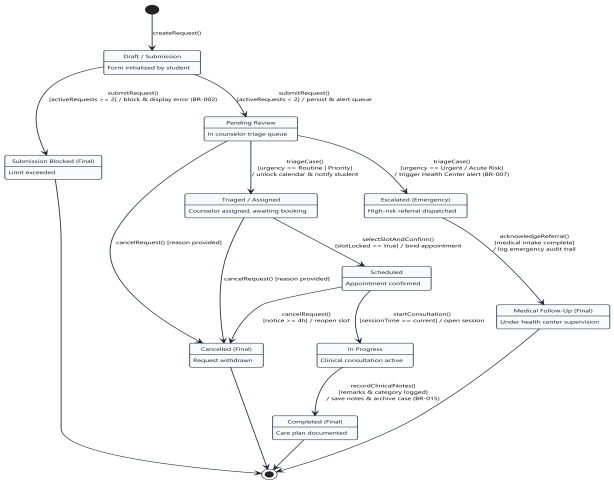


Figure 5.4: UML State Machine Diagram for ServiceRequest Entity

State Transition & Guard Condition Matrix (ServiceRequest)

Source State	Target State	Triggering Event	Guard Condition [Guard]	Action / Output
Initial [*]	Draft / Submission	`createRequest()`	None	Initialize new request form context.
Draft / Submission	Pending Review	`submitRequest()`	[activeRequests < 2]	Persist request & alert triage queue (BR-002).
Draft / Submission	Submission Blocked	`submitRequest()`	[activeRequests ≥ 2]	Block submission & display limit error (BR-002).
Pending Review	Triaged / Assigned	`triageCase()`	[urgency == Routine | Priority]	Unlock calendar & notify student.
Pending Review	Escalated (Emergency)	`triageCase()`	[urgency == Urgent / Acute Risk]	Trigger Health Center emergency alert (BR-007).
Triaged / Assigned	Scheduled	`selectSlotAndConfirm()`	[slotLocked == true]	Bind confirmed appointment ID to request.
Scheduled	In Progress	`startConsultation()`	[sessionTime == current]	Open confidential clinical session editor.
In Progress	Completed (Final)	`recordClinicalNotes()`	[remarks & category logged]	Save notes & archive case (BR-015).
Escalated (Emergency)	Medical Follow-Up (Final)	`acknowledgeReferral()`	[medical intake complete]	Log emergency resolution audit trail.
Pending / Triaged	Cancelled (Final)	`cancelRequest()`	[reason provided]	Record withdrawal reason & notify staff.

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5.2 State Machine Diagram — Entity 2: Appointment & AvailabilitySlot

The **Appointment** and **AvailabilitySlot** entities govern consultation slot publishing, atomic booking locks, rescheduling, cancellation, and session attendance tracking.

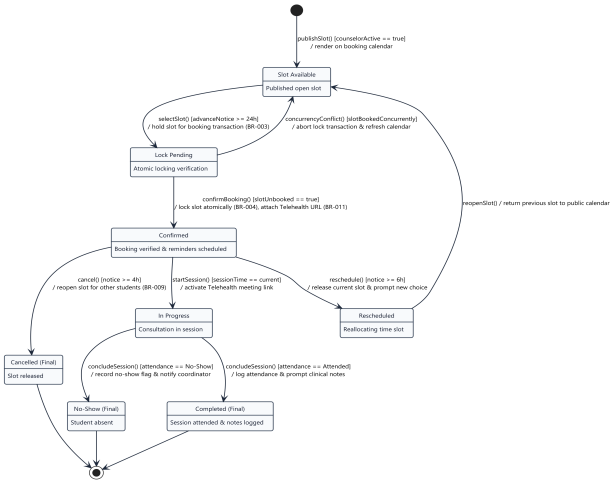


Figure 5.5: UML State Machine Diagram for Appointment & Slot Entities

State Transition & Guard Condition Matrix (Appointment & Slot)

Source State	Target State	Triggering Event	Guard Condition [Guard]	Action / Output
Initial [*]	Slot Available	`publishSlot()`	[counselorActive == true]	Render slot on student booking calendar.
Slot Available	Lock Pending	`selectSlot()`	[advanceNotice ≥ 24h]	Hold slot for booking transaction (BR-003).
Lock Pending	Confirmed	`confirmBooking()`	[slotUnbooked == true]	Lock slot atomically (BR-004), attach Telehealth URL.
Lock Pending	Slot Available	`concurrencyConflict()`	[slotBookedConcurrently]	Abort lock transaction & refresh calendar.
Confirmed	Rescheduled	`reschedule()`	[notice ≥ 6h]	Release current slot & prompt new slot choice.
Rescheduled	Slot Available	`reopenSlot()`	None	Return previous slot to public calendar.
Confirmed	Cancelled (Final)	`cancel()`	[notice ≥ 4h]	Reopen slot for other students (BR-009).
Confirmed	In Progress	`startSession()`	[sessionTime == current]	Activate virtual Telehealth meeting link.
In Progress	Completed (Final)	`concludeSession()`	[attendance == Attended]	Log attendance & prompt counselor clinical notes.
In Progress	No-Show (Final)	`concludeSession()`	[attendance == No-Show]	Record no-show flag & notify coordinator.

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Collaboration, State and Component Modelling

6. Task 3: UML Component Diagram & Subsystem Architecture

In the Student Wellness Portal, users interact via the **Client Presentation Layer** (React Web Frontend), which routes requests to the **Application & API Controller Layer** (Node.js REST Services). Services utilize **Data Access & Storage** (ORM Repositories over PostgreSQL/MySQL) and communicate with external services (LNMIIT Email SMTP Server, Telehealth URL Generator, and Campus Health Center Gateway) via standardized protocols.

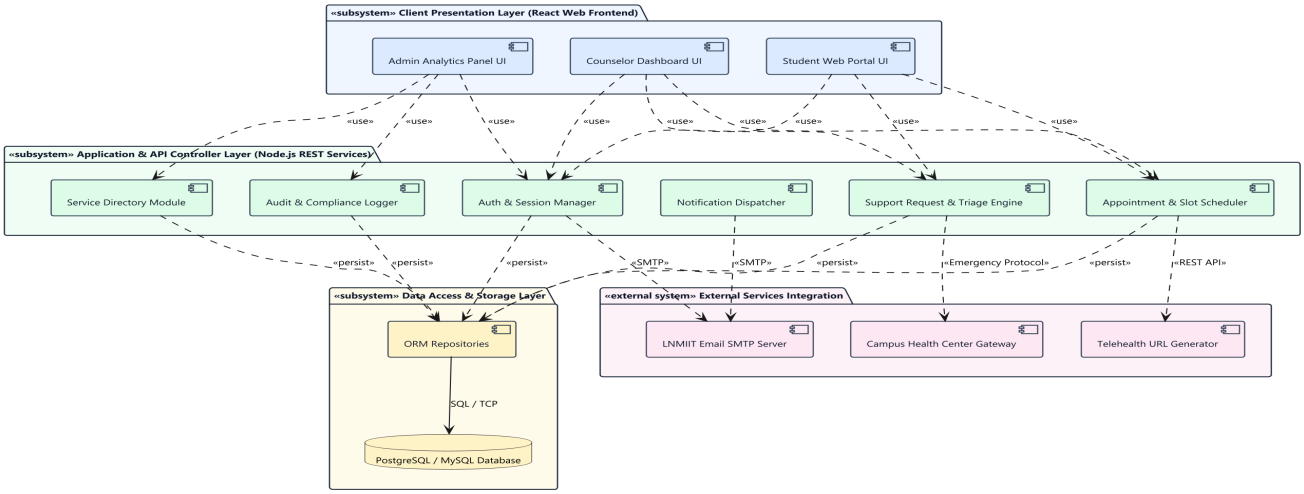


Figure 5.6: UML Component Diagram for Student Wellness Portal Architecture

Subsystem & Interface Specification Table

Subsystem	Component Name	Provided Interfaces	Required Interfaces	Dependencies
Presentation Layer	`Student Web Portal UI`	`IStudentUI`	`IAuthService, IRequestManager, ISchedulingService`	REST API endpoints
Presentation Layer	`Counselor Dashboard UI`	`ICounselorUI`	`IAuthService, ITriageEngine, ISlotLockManager`	REST API endpoints
Presentation Layer	`Admin Analytics Panel UI`	`IAdminUI`	`IAuthService, IRoleControl, IAuditLogger`	REST API endpoints
Core API Layer	`Auth & Session Manager`	`IAuthService, IRoleControl`	`IRepository, ILNMIITEmail`	bcrypt, JWT tokens
Core API Layer	`Request & Triage Engine`	`IRequestManager, ITriageEngine`	`IRepository, IEmergencyGateway`	Audit Logger, DB
Core API Layer	`Appointment Scheduler`	`ISchedulingService, ISlotLock`	`IRepository, ITelehealthGenerator`	Atomic Lock Manager
Core API Layer	`Notification Dispatcher`	`INotificationService`	`ILNMIITEmail, IWebSocketPush`	SMTP Gateway
Data Layer	`ORM Repositories`	`IRepository`	`IDatabaseEngine`	PostgreSQL / MySQL
External Services	`LNMIIT Email Server`	`ILNMIITEmail`	SMTP Protocol	@lnmiit.ac.in domain
External Services	`Telehealth URL Generator`	`ITelehealthGenerator`	Meeting REST API	Virtual URLs
External Services	`Campus Health Center`	`IEmergencyGateway`	Emergency Alert Protocol	Medical referrals

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Lab Assignment 5
Collaboration, State and Component Modelling

7. Requirements Traceability & Modelling Chain Consistency

The table below demonstrates 100% bi-directional traceability across all five lab assignments for the three carried-forward detailed use cases, verifying that no extraneous functionality or disconnected models have been introduced:

Use Case	Lab 2 Requirements	Lab 3 Activities	Lab 4 Classes & Objects	Lab 5 Collaboration	Lab 5 State Machine	Lab 5 Component
UC-005	FR-024, FR-025, FR-027, FR-028, BR-002	Multi-step request form, active limit check, Pending status.	Student, ServiceRequest, WellnessProfile, Notification	Seq 1-4: browse, profile, submit & validate limit, alert.	Draft → Pending Review (or Submission Blocked).	Request & Triage Engine, Student UI
UC-007	FR-033, FR-034, FR-035, FR-037, BR-003, BR-004, BR-011	Slot search, 24h advance check, atomic lock, Telehealth link.	Student, Counselor, AvailabilitySlot, Appointment	Seq 1-4: verify triaged, slots, lock slot & create appt, notify.	Slot Available → Lock Pending → Confirmed → Completed.	Appointment Scheduler, Telehealth Generator
UC-010	FR-029, FR-042, FR-046, BR-005, BR-007, BR-015	Triage queue evaluation, priority assignment, emergency referral.	Counselor, TriageEvaluation, HealthCenterReferral, AdminAuditLog	Seq 1-3: triage queue, evaluate, emergency referral, audit.	Pending Review → Triaged (or Escalated) → Follow-Up.	Request & Triage Engine, Health Center Gateway

8. Individual Team Contributions

Student Name & Roll No.	Individual Contribution (Lab Assignment 5)
Mehul Baid (24UCS003)	Authored UML Collaboration Diagrams and interaction tables for UC-005 (Submit Support Request) and UC-007 (Book Appointment); verified object consistency with Assignment 4 class models.
Alakh Agarwal (24UCS001)	Authored UML Collaboration Diagram for UC-010 (Conduct Triage & Case Assignment); designed UML Component Diagram and subsystem interface dependencies.
Yug Nahar (24UCS004)	Formulated UML State Machine Diagrams for ServiceRequest and Appointment entities; authored state transition matrices and guard condition tables.
Daksh Mittal (24UCS002)	Verified end-to-end modeling chain consistency across Lab Assignments 2, 3, 4, and 5; compiled document formatting; prepared final PDF submission package.

9. AI Usage Policy – Student Declaration

AI tools may be used as supporting tools for this assignment. The team is responsible for the correctness, consistency and understanding of all submitted collaboration, state and component models. We declare that: (1) We have disclosed the use of AI tools; (2) AI-generated suggestions have been reviewed, verified, edited and adapted by the team; (3) We understand and can explain all submitted models during a viva/discussion; (4) We have verified that the UML models are consistent with our project's scope, requirements, and previous assignments; (5) The final submission represents our team's understanding and modeling decisions.

Field	Declaration Details
AI Tool(s) Used:	Antigravity IDE (Gemini 3.7 Flash)
Purpose of AI Use:	Formulating UML collaboration message sequences, state machine transition matrices, component interface mapping, and aligning document formatting with course template.
Team ID:	T039
Team Member Names:	Mehul Baid (24UCS003), Alakh Agarwal (24UCS001), Yug Nahar (24UCS004), Daksh Mittal (24UCS002)
Date:	September 26, 2026
Team Confirmation:	MEHUL BAID, ALAKH AGARWAL, YUG NAHAR, DAKSH MITTAL (Signatures confirmed)